

The Broadcast Ceiling: Information and Reliability Limits of Advantage Estimators in Long-Horizon RL

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Abstract

Long-horizon language-model RL often receives terminal trajectory rewards while updating token- or step-level actions. This paper studies the information gap between those two objects. We decompose advantage-estimator error into an information ceiling and a reliability term: trajectory-constant estimators hit a broadcast ceiling when policy-relative advantages vary within a rollout, while state-, step-, token-, structural-, sampled-, or policy-implied estimators can escape that ceiling only when their information source is reliable. We evaluate this claim in CPU-only finite-MDP audits with exact behavior-policy advantages and exact policy-gradient measurements. The experiments include group-relative broadcasts, global and prefix baselines, anchor-action contrast, sampled Monte Carlo values, critic TD, an exploratory reliability-gated baseline, BRPO-style prefix diagnostics, and a VIMPO-style policy-implied signal. The new exact-gradient audit shows why token-credit correlation is not enough: REINFORCE and sibling baselines can have excellent gradient cosine while giving weak token-local explanations; critic TD has much lower gradient variance in covered states; VIMPO’s actor signal is exactly zero when $\pi_\theta = \pi_{\text{ref}}$ and becomes reference-sensitive away from that initialization. Across the 20-seed, 18-case estimator matrix, critic-style temporal-difference advantages have higher mean behavior-policy advantage correlation in 17 cases, but the phase grid also contains group-favorable and near-tie regimes. The conclusion is not that PPO universally beats GRPO. It is that estimator choice should be made on trajectory credit heterogeneity, observability, support, drift, and compute cost, not on whether a method is branded critic-based or critic-free.

Keywords. post-training; reinforcement learning from verifiable rewards; PPO; GRPO; advantage estimation; credit assignment; long-horizon agents.

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1 Introduction

Long-horizon post-training for language-model agents is not merely long-context language modeling. A trajectory may include partial plans, tool calls, retrievals, edits, failed attempts, retries, compacted summaries, and final verifier feedback. In such traces, the question “did the rollout succeed?” is not the same as “which token or action helped?” A scalar terminal reward can be useful for ranking responses, but it can also broadcast the same update over helpful subgoals, harmful detours, and delay padding.

The method question is: *for variable-length, multi-step, tool-heavy trajectories, which information source can produce reliable token- or state-varying credit under a given compute budget?* We deliberately avoid centering the work on one base model. Z.ai’s GLM-5.2 report is mentioned only as an industry example of the same design pressure: compacted, variable-length long-horizon rollouts and a move toward critic-based PPO with token-level advantages [34]. That report is not independent causal evidence that PPO beats GRPO.

Contributions. We make six contributions. First, we define the target as a behavior-policy advantage $A^{\pi_b}(x, a)$ over latent finite-MDP state and separate the information available to trajectory-constant, state-value, sampled-value, structural, prefix-verifier, and policy-implied estimators. Second, we prove an information–reliability decomposition whose trajectory-only corollary is a broadcast ceiling. Third, we introduce CPU-only finite-MDP audits with exact behavior-policy advantages and exact policy-gradient measurements. Fourth, we test stronger non-critic baselines, including anchor-action contrast, prefix-budget baselines, a BRPO-style combined baseline, and a VIMPO-style policy-implied signal. Fifth, we add phase and robustness audits over heterogeneity, observability, support, reward contrast, length imbalance, token cost, and drift. Sixth, we report downstream sanity checks: tabular closed-loop training and a tiny neural value-critic audit where the critic must interpolate to a held-out threshold.

Scope discipline. The paper makes a narrow claim about credit-assignment mechanisms. It does not claim that PPO is universally better than GRPO, that critic-free methods are trajectory-only, or that a finite-state toy predicts frontier-model behavior. We use the toy because the exact behavior-policy advantage is known, which lets us ask a question that is usually hidden in language-model training: whether an estimator is aligned with policy-relative action credit inside the trajectory. This is not the same as causal explanation. It is also not a claim that terminal-return REINFORCE is biased; REINFORCE remains an unbiased episodic policy-gradient estimator under the usual assumptions, but a trajectory-constant sample can localize action credit poorly and have high variance. This distinction matters for public research. A method can improve a benchmark for reasons unrelated to the mechanism under discussion, and a method can be mechanistically promising while still losing under a particular compute budget, implementation, reward model, or data mixture.

Paper organization. Section 2 defines the estimand and method families. Section 3 reviews related work. Section 4 gives compute and information costs. Section 5 proves the information–reliability decomposition and the broadcast ceiling. Sections 6–8 define the toy environment, experimental protocol, and results. Sections 9–11 interpret the mechanism evidence, compare method regimes, and state failure modes. The appendices provide implementation details, generated summary tables, and raw per-seed rows generated directly from the canonical JSON result artifact.

2 Definitions and Method Taxonomy

The target notation matters. Let the latent simulator state be $x_t \in \mathcal{X}$, the actor observation be $o_t = f_{\text{actor}}(x_t)$, and the critic observation be $z_t = f_{\text{critic}}(x_t)$. A sampled rollout follows a behavior policy

$$a_t \sim \pi_b(\cdot \mid o_t), \quad (1)$$

and produces a trajectory

$$\tau = (x_0, o_0, a_0, x_1, o_1, a_1, \dots, x_T). \quad (2)$$

The outcome reward $R(\tau)$ may be combined with per-token KL, process rewards, or unit-test feedback. In the finite toy below, the exact comparison target is the behavior-policy action advantage:

$$Q^{\pi_b}(x, a) = \mathbb{E}_{\pi_b}[G_t \mid x_t = x, a_t = a], \quad (3)$$

$$V^{\pi_b}(x) = \mathbb{E}_{a \sim \pi_b(\cdot \mid f_{\text{actor}}(x))} Q^{\pi_b}(x, a), \quad (4)$$

$$A^{\pi_b}(x, a) = Q^{\pi_b}(x, a) - V^{\pi_b}(x). \quad (5)$$

The superscript is intentionally not $\star$: the paper does not measure alignment with an optimal-control advantage. It measures alignment with the exact on-policy/behavior-policy advantage under the rollout distribution used to generate the batch.

2.1 PPO with a critic

PPO optimizes a clipped policy-gradient surrogate [25]:

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t [\min(\rho_t(\theta)A_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon)A_t)], \quad (6)$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)}. \quad (7)$$

In actor-critic use, a value model V_ϕ can estimate one-step temporal difference errors

$$\delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t), \quad (8)$$

or generalized advantage estimates [24]:

$$A_t^{\text{GAE}(\gamma, \lambda)} = \sum_{\ell=0}^{\infty} (\gamma\lambda)^\ell \delta_{t+\ell}. \quad (9)$$

The critic is valuable only if the value target is learnable, sufficiently covered, and cheaper than the credit it recovers.

2.2 GRPO

GRPO samples a group of completions for the same prompt and normalizes rewards inside the group [4, 26]. A simplified response-level advantage is

$$\hat{A}_i^{\text{GRPO}} = \frac{R_i - \mu_G}{\sigma_G + \epsilon}, \quad (10)$$

where μ_G and σ_G are group statistics. This critic-free design is memory-efficient and effective in many verifiable-reward settings. Its long-horizon risk is intra-rollout information loss: one scalar can be broadcast across tokens with very different policy-relative action values.

2.3 Policy-implied values

VIMPO is a crucial nearby method because it is critic-free but not trajectory-constant [9]. In a KL-regularized derivation, its actor signal can be written in a token-varying form

$$\hat{A}_t^{\text{VI}} = \beta \left[\log \frac{\pi_\theta(a_t | s_t)}{\pi_{\text{ref}}(a_t | s_t)} - D_{\text{KL}}(\pi_\theta(\cdot | s_t) \| \pi_{\text{ref}}(\cdot | s_t)) \right]. \quad (11)$$

This signal escapes any theorem that applies only to trajectory-constant estimators. It also raises a reliability question: the identity is exact at a KL-regularized optimality condition, while practical training substitutes the current policy and a reference policy. Therefore a useful testbed should ask when the policy-implied signal is aligned with the current behavior-policy advantage and when reference staleness or initialization makes it unreliable.

Table 1: Method taxonomy for long-horizon advantage signals. The important axis is whether the signal can vary within a trajectory and whether the information source is reliable. Distillation objectives are adjacent dense supervision, not direct policy-gradient estimators.

Signal source	Representative methods	Unit	Reliability question
Trajectory outcome	GRPO; RLOO; REINFORCE	resp./group	Is group reward contrast informative enough despite the broadcast ceiling?
Learned value	PPO/GAE; VAPO; ArCHer	tok./state	Is the critic calibrated under the current policy distribution?
Sampled value	VinePPO; sampled MC	state/step	Is the branching budget large enough for low-variance values?
Structural evidence	SALT; GiGPO; anchor contrast	anchor	Are repeated anchors comparable and sufficiently covered?
Prefix evidence	BRPO; process rewards	prefix/step	Are intermediate verifier calls aligned with final return?
Policy-implied value	VIMPO	tok./state	Is the reference-policy identity reliable at this training stage?
Dense distillation	OPD; OPSD; dOPSD	token/step	Does teacher or future-context supervision match the reward objective?

2.4 Variance reduction versus credit assignment

The critic in actor-critic PPO does two jobs at once. It acts as a state-dependent baseline that can reduce policy-gradient variance, and it creates token- or state-indexed temporal credit. These are different mechanisms. A running global baseline can reduce variance without telling us which step helped; a structural group method can remain critic-free while recovering some step-level contrast; a sampled value method can estimate per-state credit without training a value network.

For a single token score $\psi_t = \nabla_\theta \log \pi_\theta(a_t | s_t)$, the baseline estimator has the form

$$\hat{g}_t = (G_t - b_t)\psi_t. \quad (12)$$

Its variance is controlled by

$$\mathbb{E}[(G_t - b_t)^2 \|\psi_t\|^2], \quad (13)$$

not only by $\text{Corr}(b_t, G_t)$ or $\text{Var}(G_t - b_t)$. The variance-minimizing scalar baseline conditioned on state is

$$b^*(s_t) = \frac{\mathbb{E}[G_t \|\psi_t\|^2 | s_t]}{\mathbb{E}[\|\psi_t\|^2 | s_t]}. \quad (14)$$

BRPO is useful in this framing because its prefix-budget rewards give a critic-free, position-aware baseline source rather than a general theorem that group baselines are only variance-reduction devices [22].

Table 2: Two separable axes: how a baseline is formed and where credit is assigned. “SPO” is overloaded; here it refers separately to Single-stream Policy Optimization and Segment Policy Optimization.

Baseline source	Trajectory-level credit	Step-/segment-level credit
None	REINFORCE	–
Global	REINFORCE++; Single-stream PO	add process rewards
Sibling/LOO	GRPO; RLOO; DAPO; Dr. GRPO	GiGPO; SALT; anchor contrast; PRM+GRPO
Learned critic $V(s)$	usually unnecessary alone	PPO/GAE; ArCHer
Sampled value	–	VinePPO; OPPO; Segment PO
Policy-implied	–	VIMPO
Redistribution	–	RUDDER-style decomposition

2.5 Exploratory reliability-gated baseline

The counterexample in this paper suggests a simple adaptive estimator. A critic should not be used merely because a value model exists; it should be used where the value estimate is locally supported. Let $n_D(s_t)$ be a replay or batch coverage count for the critic state abstraction, k a minimum support threshold, $\hat{A}_t^{\text{TD}}$ a critic temporal-difference advantage, and $\hat{A}_i^G$ the group-relative scalar for trajectory i . Define

$$m_t = \mathbf{1}\{n_D(s_t) \geq k\}, \quad \hat{A}_t^{\text{CG}} = m_t \hat{A}_t^{\text{TD}} + (1 - m_t) \hat{A}_i^G. \quad (15)$$

This reliability-gated estimator is intentionally modest. It does not solve critic blindness: if the state abstraction hides the causal feature, high coverage can still be misleading. Its purpose is to make critic readiness an explicit experimental variable and to preserve group-relative signal where the critic is locally undercovered.

3 Related Work

PPO and actor–critic post-training. PPO was introduced as a simpler clipped-objective alternative to trust-region policy optimization [25]. In RLHF-style language-model training, actor–critic optimization is usually wrapped with KL penalties, reference policies, reward models, old-logprob storage, reward normalization, and sequence-level preference signals [21, 27]. The relevant point for this work is not one historical RLHF recipe, but the availability of a state- or token-indexed value estimate. A critic can turn delayed outcome feedback into temporally structured advantages. That can help when rollouts contain subgoals and detours, but it also creates a second model that can drift, overfit, or assign confident false credit.

GRPO and RL with verifiable rewards. DeepSeekMath introduced GRPO as a critic-free method for mathematical reasoning, replacing the learned value model with group-relative reward normalization [26]. DeepSeek-R1 then made GRPO-style RL a central public reference point for reasoning-model training [4]. Subsequent work has analyzed the effective loss, success amplification, response-length effects, dynamic sampling, clipping details, group-relative bias, and statistical structure of GRPO-like gradients [17, 20, 23, 30, 31, 37]. The practical attraction is real: GRPO removes critic memory and critic training, which can be decisive at large scale. The risk studied here is that a response-level group scalar can be information-wasteful inside long, heterogeneous rollouts.

Baselines, sampled values, and step credit. Recent work shows that the design space is broader than PPO versus GRPO. REINFORCE-style and global-normalization methods reduce critic cost without necessarily solving intra-rollout credit [2, 8, 29]. VAPO is a particularly close value-based reasoning reference: it argues that long-CoT training needs reliable value estimation while explicitly addressing value bias, heterogeneous lengths, and sparse rewards [33]. VinePPO estimates values by branching Monte Carlo rollouts rather than fitting a value network [10]. ArCher uses a hierarchical multi-turn value structure for language-model agents [38]. OPPO is a very recent token-credit proposal based on Bayesian value recursion and should be read as a promising taxonomy anchor rather than settled evidence [14]. RUDDER and Segment Policy Optimization connect the same question to delayed-reward redistribution and segment-level credit [3, 7].

Dense and structural step signals. Process supervision is another way to avoid a pure terminal broadcast. Stepwise verification and process reward models provide direct intermediate feedback [16, 28], while implicit process rewards try to recover denser signals from cheaper outcome supervision [32]. GiGPO and SALT are especially relevant because they preserve critic-free or group-based structure while constructing finer-grained comparisons across repeated states or trajectory graphs [5, 12]. These methods make the main point sharper: “critic-free” does not have to mean “trajectory-only,” and “critic-based” does not automatically mean correct credit. Our anchor-action contrast row is a minimal local analogue of that structural idea for the toy environment; it is not a reproduction of GiGPO or SALT. Evidence-Calibrated Policy Optimization is especially close to our anchor-coverage and gating discussion: it shows that repeated-anchor evidence can be unreliable under limited rollout support and uses count shrinkage plus variance-gated critic-free credit on ALFWorld and WebShop [15]. Our novelty is therefore narrower: a broadcast-ceiling diagnostic, a finite-MDP phase diagram, and a cautious adaptive-selection framing.

Policy-implied and prefix-budget signals. VIMPO is the closest recent method to the taxonomy proposed here because it is critic-free while producing a token-varying signal from policy/reference log-ratios and a KL-regularized value identity [9]. Its existence is one reason this paper avoids the binary phrase “critic versus critic-free” as the main axis. The finite-MDP audit below treats VIMPO as a policy-implied estimator whose reliability depends on reference consistency and distance from the optimality condition. AnytimeReasoner and BRPO are also important because they use verifier calls at multiple thinking budgets to construct prefix-conditioned baselines [22]. We use their insight to report score-weighted residual-return diagnostics, while not claiming that a small MDP reproduces their full anytime-reasoning setup.

On-policy distillation. OPD and related generalized knowledge distillation methods train on student-visited states, reducing train/inference mismatch relative to purely offline teacher forcing [1, 13]. OPSD uses the same model in privileged teacher and ordinary student roles, aiming to capture dense reasoning supervision without a separate teacher model [36]. Recent OPD analyses emphasize teacher/student compatibility, teacher novelty, length inflation, and truncation collapse [13, 18]. These methods are close to the present problem because they also produce dense token-level signals on rollouts. They differ because their signal is a distillation target rather than a reward-improvement objective. Recent work extends on-policy self-distillation to diffusion LLMs. d-OPSD constructs a suffix-conditioned self-teacher from the model’s own completed generation and applies divergence supervision at the denoising-step level, illustrating that the natural unit of dense supervision depends on the generation architecture [19]. Because this is a distillation objective rather than an advantage estimator, we treat it as adjacent rather than directly comparable work.

Long-horizon agentic training. Long-horizon agentic RL introduces issues that are often absent in short verifier tasks: tool-call state, file-system state, hidden evaluation artifacts, environment side effects, and compaction. The GLM-5.2 report is useful because it explicitly names compacted sub-traces, variable numbers of trainable traces per prompt, and a move from group-wise optimization to critic-based PPO with token-level advantages [34]. We treat this as a case study of a design pressure, not as proof of a causal performance mechanism. The 2026 credit-assignment survey [35] is useful context for why this paper is scoped as a diagnostic rather than a universal method claim: long-horizon reasoning and agentic RL span token, segment, step, turn, counterfactual, value-based, process-reward, and graph-based mechanisms.

4 Compute and Information Costs

Fair comparisons should charge all hidden inputs, not only final benchmark score. For PPO, the bill includes value-model parameters, value targets, critic forward passes, optimizer state, old-policy log probabilities, KL control, and rollout storage. For GRPO, the bill includes group size, discarded completions, low-dispersion groups, and verifier calls. For VIMPO, it includes reference-policy scoring and full-action KL approximations. For prefix-verifier or BRPO-style methods, it includes intermediate verifier calls. For structural methods, it includes matching repeated anchors or trajectory graphs. For dense distillation, it includes teacher or privileged-context compute, dense logits, trajectory length inflation, and truncation handling.

The near-term reporting scorecard for this workstream is the reward change reported beside its resource vector, not a literal scalar objective:

$$\Delta \text{reward} \quad \text{given} \quad (\text{generated tokens, wall clock, extra model memory}), \quad (16)$$

reported alongside reproducible artifacts and negative results.

Policy and rollout cost. For PPO, each trainable token typically needs old-policy log probabilities, advantages, reference-policy information for KL control, and value-model targets. If rollouts are compacted into sub-traces, the implementation must decide whether every sub-trace is trainable, how truncation is handled, and how losses are normalized across very different lengths. GRPO avoids the critic but spends compute on group sampling. If many group members have identical verifier outcomes, the effective signal can be weak even when token generation cost is high.

Synchronization and straggler cost. Group-relative methods also have systems costs. AReaL documents the underutilization that arises when synchronous large-scale RL waits for long rollouts, and studies asynchronous training as a throughput/staleness tradeoff [6]. Straggler-aware group sizing studies the same pressure in synchronous GRPO-like training: larger groups can improve statistical baselines but increase the wall-clock risk that one long rollout stalls the group [11]. These are compute-cost issues, not direct credit-assignment theorems.

Distillation cost. OPD and OPSD can look cheap if only student updates are counted. A fair bill also includes teacher logits or hidden states, privileged traces, context construction, rejection/truncation policies, and any additional sampling needed to keep the student on-policy. Dense token supervision can be worth this cost, but it should not be compared to RL methods as if the dense signal were free.

5 Formal Limitation: Information, Reliability, and Broadcast

The title of this paper should not be read as “terminal-return policy gradients are biased.” A trajectory-level return can be a valid Monte Carlo coefficient for the episodic policy gradient:

$$\nabla J(\theta) = \mathbb{E}_{\tau \sim \pi_\theta} \left[R(\tau) \sum_t \nabla_\theta \log \pi_\theta(a_t | o_t) \right]. \quad (17)$$

REINFORCE remains an unbiased estimator of this gradient under the usual on-policy assumptions. The issue studied here is different: a trajectory-constant scalar can be a poor sample-level localizer of policy-relative action credit inside a long heterogeneous rollout. It can reinforce helpful, harmful, and delay actions together in a single sampled trace, even though the expectation over many score vectors may still be correct for the total gradient.

Information–reliability decomposition. Let

$$Y = A^{\pi_b}(x_t, a_t) \quad (18)$$

be the target advantage for a sampled token. Let Z_m denote all information available to estimator family m : a trajectory return, sibling rollouts, critic state, sampled continuations, repeated anchors, prefix verifiers, or policy/reference log-ratios. For any estimator $\hat{A}_m = h_m(Z_m)$,

$$\mathbb{E}[(Y - \hat{A}_m)^2] = \underbrace{\mathbb{E}[\text{Var}(Y | Z_m)]}_{\text{information ceiling}} + \underbrace{\mathbb{E}[(\mathbb{E}[Y | Z_m] - h_m(Z_m))^2]}_{\text{reliability or consistency error}} \quad (19)$$

by adding and subtracting $\mathbb{E}[Y | Z_m]$. The best possible squared correlation using the information source Z_m is

$$\max_h \text{Corr}(Y, h(Z_m))^2 = \frac{\text{Var}(\mathbb{E}[Y | Z_m])}{\text{Var}(Y)}. \quad (20)$$

This is the core decision rule. A method helps when it reduces information error more than it increases reliability error and compute:

$$\hat{m} = \arg \min_m \left\{ \hat{\mathcal{I}}_m + \hat{\mathcal{E}}_m + \lambda \text{Cost}_m \right\}. \quad (21)$$

Trajectory-constant methods have a large $\hat{\mathcal{I}}$ when within-rollout credit is heterogeneous. Critics have smaller information error when state is sufficient, but can have large fitting or calibration error. VIMPO has token-varying information, but its reliability depends on reference consistency and the KL-regularized optimality identity. Structural methods depend on repeated comparable states. Prefix methods depend on intermediate verifier quality.

Trajectory-broadcast ceiling. Sample a token pair (τ, t) according to the token weighting used for training, and define

$$Y = A^{\pi_b}(x_t, a_t). \quad (22)$$

Consider all estimators that are constant within a trajectory,

$$\hat{A}_t = g(\tau). \quad (23)$$

The L_2 -optimal member of this class is

$$g^*(\tau) = \mathbb{E}[Y | \tau], \quad (24)$$

and its irreducible error is

$$\min_g \mathbb{E}[(Y - g(\tau))^2] = \mathbb{E}[\text{Var}(Y \mid \tau)]. \quad (25)$$

By the law of total variance,

$$\max_g \text{Corr}(Y, g(\tau))^2 = \frac{\text{Var}(\mathbb{E}[Y \mid \tau])}{\text{Var}(Y)} = 1 - \frac{\mathbb{E}[\text{Var}(Y \mid \tau)]}{\text{Var}(Y)}. \quad (26)$$

Define the credit heterogeneity index

$$H_{\text{credit}} = \frac{\mathbb{E}[\text{Var}(A_t^{\pi_b} \mid \tau)]}{\text{Var}(A_t^{\pi_b})}. \quad (27)$$

Then any trajectory-constant estimator has absolute token-level correlation at most

$$|\text{Corr}(A_t^{\pi_b}, g(\tau))| \leq \sqrt{1 - H_{\text{credit}}}. \quad (28)$$

This trajectory-broadcast ceiling does not say that group or global methods cannot produce useful policy-gradient updates. It says that, when H_{credit} is large, any method that emits one scalar per trajectory has an information-theoretic token-fidelity ceiling. Learned critics, sampled values, process rewards, or structural repeated-state methods can beat that ceiling only if their own held-out reliability is high enough.

Operational diagnostic. The finite-MDP experiments report descriptive quantities side by side:

$$H_{\text{credit}} \quad \text{and} \quad R_{\text{critic}} = \text{Corr}(\hat{A}_t^V, A_t^{\pi_b}) \quad (29)$$

on held-out toy rollouts. Because R_{critic} directly uses the oracle target, it is not presented as an oracle-free deployment diagnostic. In real LLM training, H_{credit} should be estimated by a small branching audit, process labels, sampled values, or counterfactual rollouts. Critic reliability should be estimated by held-out value MSE, calibration slope/intercept, Bellman residuals, bootstrap or ensemble disagreement, policy-distribution drift, state-support counts, and rank correlation across independent continuations. The practical selection question is whether these observable proxies predict lower held-out estimator-selection regret.

6 Toy Experiment

6.1 Environment

The toy environment creates variable-length trajectories with three action types: **help**, **harm**, and **wait**. **help** increases a hidden progress score, **harm** decreases it, and **wait** leaves progress unchanged while consuming one decision step. It is therefore a *delay* action, not a true null action: even with zero explicit token cost, it can reduce future opportunity. A terminal verifier returns success when the final score reaches a prompt-specific threshold. Because the finite dynamics and behavior policy are known, we can compute the exact behavior-policy advantage:

$$A^{\pi_b}(x_t, a_t) = -c + V^{\pi_b}(x_{t+1}, h - 1) - V^{\pi_b}(x_t, h), \quad (30)$$

where c is token cost and h is remaining horizon. Result artifacts keep the legacy field `oracle_advantage` for backward compatibility, but the estimand is A^{π_b} , not an optimal value advantage.

The exact-gradient audit uses a closely related finite MDP with four actions: **help**, **harm**, **delay**, and **null**. The **delay** action consumes effective decision horizon. The **null** action is a true irrelevant emission: it changes neither progress nor effective future opportunity. Its Bellman contribution is solved as a zero-reward self-loop, so its exact one-step advantage is zero. This lets the audit measure whether an estimator leaks credit onto irrelevant tokens separately from whether it penalizes opportunity-consuming delays.

6.2 Estimators and metrics

The group-relative estimator normalizes terminal returns inside each prompt group and broadcasts the resulting scalar to all tokens in the trajectory. The critic-style estimator fits a tabular value model from sampled returns-to-go and computes one-step temporal-difference advantages. We measure Pearson correlation with A^{π_b} , calibrated MSE after affine rescaling, raw MSE, sign accuracy, delay-action magnitude, within-trajectory variance, zero-variance group fraction, and exact critic-state hit rate.

The variance-credit grid adds eight non-exact estimators on the same toy dynamics: raw REINFORCE returns, a global return baseline, sibling group normalization, leave-one-out group centering, a dense progress-reward proxy, a critic-free anchor-action contrast over repeated state-action visits, a learned critic TD estimator, and a sampled Monte Carlo value estimator. The purpose is not to simulate every cited algorithm, but to test the mechanism classes in Table 2: baseline-only variance reduction, structural critic-free step credit, and value-estimation-based step credit.

Three robustness audits test likely objections. The anchor-coverage audit sweeps evaluation batch size to change how often exact state-action anchors repeat, testing whether the structural critic-free baseline is coverage-bound. The length-imbalance audit adds a length-adjusted group scalar, computed by normalizing return per token inside each prompt group, and sweeps maximum horizon. The token-cost audit reruns baseline and long-wait settings with per-token costs $c \in \{0, 0.01, 0.02, 0.05\}$, including the no-explicit-length-penalty case $c = 0$.

The phase-diagram audit uses the same environment but sweeps credit heterogeneity, critic observability, critic coverage, reward contrast, and policy drift. For each cell it reports H_{credit} , the implied trajectory-broadcast ceiling, group correlation, critic reliability, and a crossover reading. This replaces a winner-count-only narrative with a method-selection diagnostic.

The closed-loop training audit uses the same toy dynamics but trains a tabular softmax policy with policy-gradient updates under matched generated-token budgets. This is still not full PPO or production GRPO: there is no neural policy, KL controller, minibatch optimizer, or clipped surrogate. It does test whether estimator-fidelity differences survive one downstream optimization loop. The compared signals are group total return, group return per token, critic TD from a replay value table, and the exploratory reliability-gated baseline.

The neural generalization audit keeps the policy fixed and replaces the tabular critic with a one-hidden-layer value network trained by SGD on thresholds 1 and 3, then evaluated on held-out threshold 2. Exact tabular state lookup cannot help in this setting because threshold is part of the exact state key. The audit therefore asks whether a small value model can interpolate enough structure to recover token-level temporal credit.

The behavior-policy-alignment matrix remains an estimator-fidelity experiment under controlled dynamics. The closed-loop audit is a downstream tabular-policy stress test, not a production PPO-vs-GRPO benchmark.

Exact policy-gradient audit. Because the MDP is finite, the exact-gradient variant computes the expected return $J(\theta)$ and a numerical exact policy gradient $g^* = \nabla J$. For each estimator, we

sample batches, form $\hat{g} = \sum_t \hat{A}_t \nabla_{\theta} \log \pi(a_t | s_t)$, and report relative gradient bias, trace variance, cosine with g^* , and one-step improvement under a matched policy-KL step. This audit also includes VIMPO-style policy-implied advantages, sibling leave-one-out baselines, prefix-budget baselines, a BRPO-style combined baseline, and critic TD.

7 Experimental Protocol

Design principle. The experiment is designed around a known behavior-policy target. In real language-model RL, we usually observe a reward and a model update but not the policy-relative action advantage. Here the finite-state dynamics allow exact dynamic programming for V^{π_b} , which turns the question into an estimator-fidelity problem. That lets us separate two claims that are often conflated: whether a method is cheap and whether its advantage signal points at the right actions under the rollout policy.

Case axes. The deep matrix varies five axes. The horizon axis tests whether scalar terminal broadcasts become less aligned as trajectories lengthen. The group-size axis tests whether more completions per prompt rescue GRPO-like normalization. The critic-budget axis tests coverage: a value table trained from too few groups should not be trusted. The observability axis removes or coarsens state features to simulate poor value-model inputs. The sparse-reward axis tests harder terminal feedback. Finally, the counterexample intentionally combines a blind critic and low coverage to create a regime where group-relative terminal outcomes should win.

Decision rule. The headline statistic is Pearson correlation with A^{π_b} , treated as a secondary interpretability metric rather than a full policy-gradient fidelity measure. A case is a mean critic win when the mean critic correlation exceeds the mean group-relative correlation. Because one mean win can be tiny, we also report a 95% across-seed confidence interval for the delta. A “clear critic” case has positive delta after subtracting the interval; a “clear group” case has negative delta after adding the interval; otherwise the case is labeled a near tie. The phase diagnostic uses an additional practical band of $|\Delta r| < 0.03$ before calling a crossover clear. This is why the abstract reports both the mean-win count and the more cautious confidence-interval reading.

Why not a full closed-loop PPO/GRPO benchmark? The closed-loop audit below is intentionally tabular and unclipped. A full PPO/GRPO benchmark would mix estimator fidelity with optimizer settings, clipping schedules, policy parameterization, exploration, and KL control. This paper therefore asks the behavior-policy credit question first, then uses a small downstream policy-training audit as a sanity check rather than as final production evidence.

8 Results

The canonical matrix uses 20 seeds and 18 fixed cases across horizon length, group size, critic budget, observability, sparse-reward regimes, and a designed counterexample. Critic-style TD has the higher mean correlation in 17 of 18 cases. A confidence-interval reading is more cautious: 16 cases are clearly critic-favorable, 1 is a near tie, and 1 is clearly group-favorable. The mean critic-minus-group correlation is 0.420. Table 3 reports representative cases; the full CSV and JSON are committed in `results/`.

Critic minus group correlation by case

Positive values favor critic-style TD; negative values favor group-relative advantage.

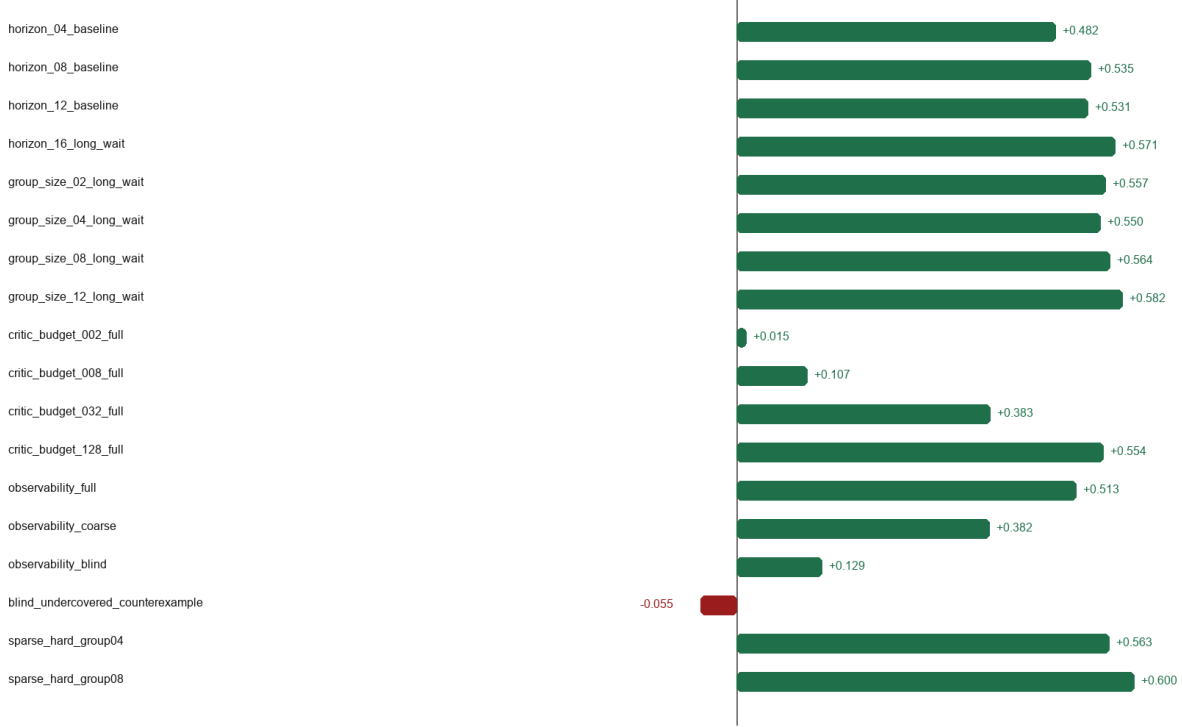


Figure 1: Mean critic-minus-group behavior-policy advantage correlation by case. Positive values favor critic-style temporal-difference advantages.

Table 3: Selected rows from the 20-seed deep matrix. Delta is critic r minus group r ; CI is a 95% across-seed interval for delta.

Case	Group r	Critic r	Delta	CI	Reading
horizon_04_baseline	0.495	0.977	0.482	± 0.011	critic clear
horizon_16_long_wait	0.293	0.865	0.571	± 0.013	critic clear
critic_budget_002_full	0.352	0.367	0.015	± 0.027	near tie
critic_budget_128_full	0.352	0.906	0.554	± 0.017	critic clear
observability_blind	0.353	0.482	0.129	± 0.017	critic clear
blind_undercovered_counterexample	0.510	0.455	-0.055	± 0.015	group clear
sparse_hard_group08	0.310	0.910	0.600	± 0.008	critic clear

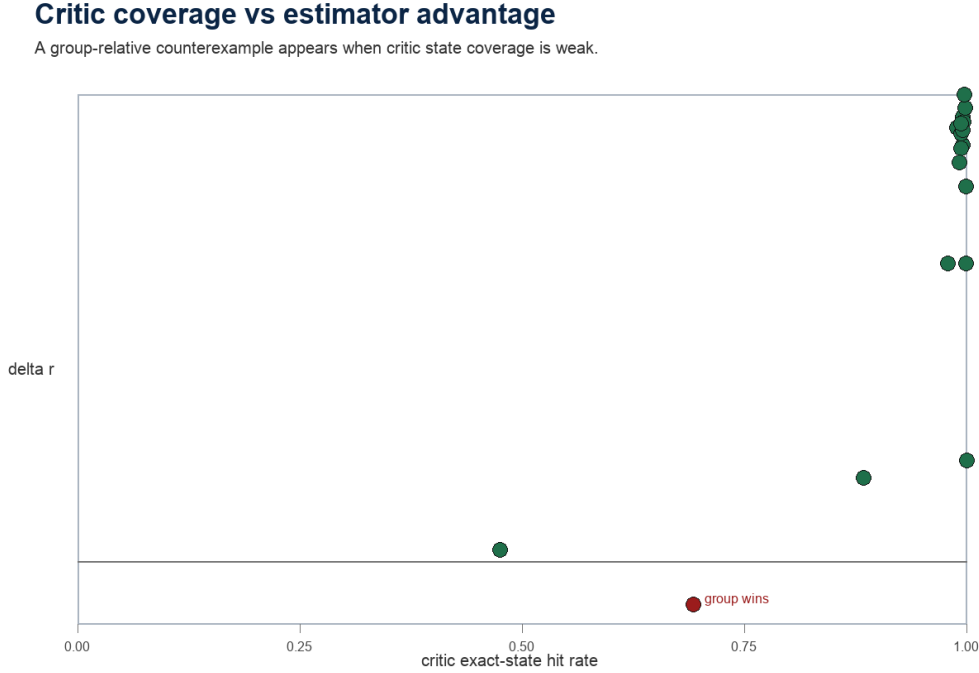


Figure 2: Critic state coverage versus estimator advantage. The group-relative counterexample occurs when the critic is blind and undercovered.

Interpretation. Within this controlled estimator-fidelity toy, the evidence supports the PPO-favorable hypothesis when progress state is observable and critic coverage is adequate: critic-style advantages track policy-relative action credit more closely than response-level group scalars. The result also protects against overstatement. In the coverage-limited case, the confidence interval crosses zero; in the blind and undercovered case, group-relative terminal outcome information is better than a poor value table.

Table 4: Exact finite-MDP policy-gradient fidelity. Bias is $\|\mathbb{E}[\hat{g}] - g^*\|/\|g^*\|$, Var. is the across-batch trace variance, Cos. is gradient cosine with g^* , and ΔJ is exact return improvement after a matched policy-KL step. r_A is token-level advantage correlation, included as a secondary diagnostic.

Estimator	Bias/ $\ g^*\ $	Var.	Cos.	ΔJ	r_A	Cal. MSE	Reading
REINFORCE	0.108	0.022	0.994	0.190	0.430	0.0365	unbiased direction but high variance
Sibling LOO	0.071	0.017	0.998	0.191	0.386	0.0381	lower variance group baseline
Prefix baseline	0.061	0.013	0.998	0.191	0.429	0.0365	coarse prefix/budget baseline
BRPO-style	0.063	0.014	0.998	0.191	0.416	0.0370	combines sibling and prefix baselines
Critic TD	0.028	0.003	1.000	0.192	1.000	0.0000	lowest variance in this supported tabular setting
VIMPO, $\pi = \pi_{\text{ref}}$	1.000	0.000	0.000	0.000	0.000	0.0448	zero actor signal at initialization
VIMPO, stale ref.	0.750	0.001	0.661	0.125	0.583	0.0296	token-varying but reference-sensitive

$\|g^*\| = 0.160$; base return 0.506; null-token fraction 0.072; stale-reference KL 0.040.

Exact-gradient audit. Tables 4 and 5 are the main upgrade over correlation-only evidence. REINFORCE and sibling leave-one-out have excellent gradient cosine with the exact policy gradient

Table 5: Prefix and baseline diagnostics from the exact-gradient audit. The score-weighted residual moment is $\mathbb{E}[(G_t - b_t)^2 \|\nabla \log \pi(a_t | s_t)\|^2]$, the quantity most directly related to single-token score-function variance.

Baseline	$\text{Corr}(b_t, G_t)$	$\text{Var}(G - b)/\text{Var}(G)$	Score-wtd. resid.	Count
Group mean	-0.016	1.260	0.195	29095
Prefix budget	0.053	1.000	0.156	29095
BRPO-style	-0.015	1.069	0.166	29095
Critic value	0.457	0.791	0.128	29095

even though their token-level advantage correlations are modest. That is expected: terminal returns can be valid Monte Carlo policy-gradient coefficients while still being weak local credit explanations. Critic TD has the lowest trace variance in this covered tabular setting. The VIMPO-style equal-reference row is the revealing sanity check: when $\pi_\theta = \pi_{\text{ref}}$, its actor signal is exactly zero even though the exact behavior-policy gradient is nonzero. With a stale reference, the signal becomes token-varying and partially aligned ($\cos = 0.661$), but remains biased relative to the current behavior-policy advantage. The baseline table gives the analogous BRPO-style message: prefix and combined baselines can reduce score-weighted residual moments relative to a group mean, but the critic value baseline remains strongest in this supported toy.

Table 6: Broadcast-ceiling phase diagnostic over credit heterogeneity, observability, critic coverage, reward contrast, and policy drift. The ceiling is $\sqrt{1 - H_{\text{credit}}}$, the best possible absolute correlation for a trajectory-constant estimator.

Regime	H_{credit}	Ceiling	Group r	Critic r	Δr	Crossover reading
low full high contrast matched	0.671	0.573	0.457	0.968	0.511	critic clear; critic/sampled value
low blind low sparse matched	0.743	0.506	0.364	0.309	-0.055	group clear; process/structural/hybrid
high full high contrast matched	0.845	0.393	0.277	0.826	0.549	critic clear; critic/sampled value
high full low sparse stale	0.879	0.347	0.283	0.245	-0.037	near tie; process/structural/hybrid
high blind high contrast matched	0.845	0.393	0.277	0.408	0.132	critic clear; process/structural/hybrid
high blind low sparse stale	0.879	0.347	0.283	0.235	-0.048	near tie; process/structural/hybrid
32 cells; 15 critic clear, 2 group clear, 15 near ties.						

Broadcast-ceiling phase diagram. Table 6 is the main response to the boundary critique. In the 32-cell phase grid, high-support full-state critics produce clear critic-favorable crossovers, while blind low-support sparse cells produce clear group-favorable crossovers. The diagnostic also shows many near ties, which is useful: high H_{credit} lowers the ceiling for trajectory-constant estimators, but a weak critic does not automatically exploit that opening. The stronger working claim is therefore conditional: critic benefit is predicted jointly by within-trajectory credit heterogeneity and held-out critic reliability.

Table 7: Variance-reduction versus credit-assignment grid on the long-wait toy case. Baseline-only methods can lower second moment, but they do not create within-trajectory credit variation. Abbreviations: Var. is variance reduction; Group z is sibling-group z-scoring; LOO is leave-one-out; Within is within-trajectory estimator variance; Anchor ctr. is anchor-action contrast.

Estimator	Var.	Credit	r	MSE	Wait	Within
REINFORCE	none	traj.	0.357	0.0208	1.157	0.000
Global	global	traj.	0.357	0.0208	0.875	0.000
Group z	group	traj.	0.319	0.0214	0.855	0.000
LOO	group	traj.	0.332	0.0212	0.799	0.000
Dense reward	dense	step	0.674	0.0130	0.000	0.416
Anchor ctr.	anchor	step	0.715	0.0117	0.439	0.042
Critic TD	critic	step	0.870	0.0058	0.356	0.031
Sampled MC	sampled	step	0.849	0.0067	0.563	0.034

Variance-credit decomposition. Table 7 makes the decomposition explicit. The global baseline reduces the second moment of the trajectory-level policy gradient signal relative to raw REINFORCE returns, but it leaves within-trajectory variance at zero because every token in a trajectory still receives the same scalar. Sibling group normalization is memory-light and critic-free, but in this long-wait setting it also remains a trajectory-level broadcast. The anchor-action contrast row tests a stronger critic-free structural baseline: it uses repeated state-action anchors to create local step-level comparisons without fitting a value model. It is a transductive batch estimator over exact repeated toy states: it assumes repeated state-action coverage, and unsupported anchors receive zero rather than a borrowed value estimate. It substantially narrows the gap to the critic rows, improving correlation from the sibling-group baseline’s 0.319 to 0.715, but it remains below learned critic TD at 0.870 and sampled Monte Carlo value estimation at 0.849. The dense progress-reward proxy also gives a step signal, but because it lacks horizon and threshold information, it is not a full value estimate. The strongest claim supported by this grid is therefore not that every critic-free method is weak; it is that long-horizon tasks need some mechanism for state-sensitive step credit, and in this toy the value-estimation mechanisms still carry the highest behavior-target alignment.

Table 8: Anchor-coverage audit over five seeds. Groups is the number of evaluation prompt groups in the estimator batch; Support is the fraction of steps with leave-one-out state-action anchor support. $A - G$ is anchor r minus group r ; $C - A$ is critic r minus anchor r .

Groups	Support	Group r	Anchor r	Critic r	$A - G$	$C - A$
2	0.413	0.361	0.134	0.805	-0.228	0.672
4	0.510	0.341	0.253	0.836	-0.088	0.583
8	0.675	0.329	0.389	0.852	0.060	0.463
16	0.813	0.326	0.544	0.845	0.219	0.301
32	0.915	0.321	0.654	0.861	0.334	0.206
48	0.945	0.318	0.710	0.867	0.392	0.156
64	0.962	0.315	0.751	0.865	0.437	0.113

Anchor coverage audit. Table 8 checks the assumption behind the structural critic-free baseline. At two evaluation groups, only 41.3% of steps have supported repeated anchors and anchor contrast is worse than sibling grouping ($r = 0.134$ versus $r = 0.361$). The method crosses the sibling baseline at eight groups, reaches support above 0.80 at 16 groups, and climbs to $r = 0.751$ at 64 groups. Critic TD remains higher in every row. This is a useful negative and positive result at once: structural critic-free step credit can be real, but it is not free. It needs repeated comparable states,

and sparse-anchor regimes should fall back to group or critic signals rather than pretending a local contrast exists.

Table 9: Length-imbalance audit on the long-wait scenario. Len gap is the mean within-group max-minus-min trajectory length. Len-adj. is a group scalar based on return per token; Δ_G and Δ_L are critic r minus group-total and length-adjusted group r .

$H_{\max}$	Len gap	Wait	Group r	Len-adj. r	Critic r	Δ_G	Δ_L
4	1.775	0.336	0.488	0.500	0.981	0.492	0.481
8	4.851	0.431	0.389	0.389	0.940	0.551	0.551
12	7.725	0.495	0.330	0.328	0.890	0.561	0.562
16	10.696	0.539	0.293	0.289	0.849	0.556	0.560
20	13.439	0.573	0.266	0.260	0.811	0.545	0.551

Length-imbalance audit. Table 9 tests whether a simple length correction rescues trajectory-level grouping. It does not in this toy. As $H_{\max}$ increases from 4 to 20, mean within-group length range grows from 1.775 to 13.439 tokens and delay-action fraction grows from 0.336 to 0.573. The length-adjusted group scalar remains close to the total-return group scalar: at $H_{\max} = 20$, critic r is 0.811, group-total r is 0.266, and length-adjusted group r is 0.260. Length correction changes the rollout ranking signal, but it still broadcasts one number to all tokens.

Table 10: Token-cost sensitivity over 20 seeds. The $c = 0$ rows remove the explicit per-token penalty. CI is a 95% interval for critic-minus-group correlation; Wait_G and Wait_C are delay-action magnitude ratios.

Scenario	Cost	Group r	Critic r	Δ	CI	Wait_G	Wait_C
base	0.00	0.336	0.882	0.546	0.010	0.930	0.465
base	0.01	0.332	0.882	0.550	0.010	0.937	0.465
base	0.02	0.335	0.882	0.546	0.010	0.928	0.465
base	0.05	0.339	0.882	0.543	0.010	0.902	0.465
long	0.00	0.335	0.888	0.552	0.015	0.871	0.371
long	0.01	0.327	0.888	0.561	0.016	0.911	0.371
long	0.02	0.330	0.888	0.557	0.016	0.898	0.371
long	0.05	0.334	0.888	0.554	0.016	0.868	0.371

Token-cost sensitivity. Table 10 checks whether the critic result is just a side effect of the per-token cost. The answer is no for these controlled dynamics. All eight scenario/cost rows have positive critic-minus-group correlation after subtracting the 95% interval. In the long-wait, zero-cost row, critic r is 0.888, group r is 0.335, and the delta is 0.552. Removing the explicit length penalty therefore does not remove the temporal-credit gap.

Closed-loop training. Table 11 is the first downstream optimization check. The result is more conservative than the estimator-fidelity tables, as expected. In the default setting, group-total training improves mean return from 0.515 to 0.785, while critic TD reaches 0.810 and the reliability-gated baseline reaches 0.809. In the low-coverage stress setting, group total reaches 0.769, critic TD reaches 0.780, and the reliability-gated baseline reaches 0.777 while using critic credit on only 64.4% of final-batch steps. The adaptive estimator is therefore a candidate design pattern, not a claimed replacement for PPO or GRPO: it preserves most critic benefit while making critic use explicit and auditable.

Table 11: Closed-loop tabular policy training under matched generated-token budgets. Default uses the standard replay budget; low cov. restricts replay and raises the reliability gate. Cov-gated is an exploratory reliability-gated baseline.

Setting	Method	Init R	Final R	ΔR	Success	Critic frac.
default	Group	0.515	0.785	0.271	0.949	0.000
default	Len-group	0.515	0.782	0.267	0.945	0.000
default	Critic TD	0.515	0.810	0.296	0.974	0.998
default	Cov-gated	0.515	0.809	0.294	0.972	0.996
low cov.	Group	0.508	0.769	0.261	0.928	0.000
low cov.	Len-group	0.508	0.754	0.246	0.913	0.000
low cov.	Critic TD	0.508	0.780	0.272	0.940	0.945
low cov.	Cov-gated	0.508	0.777	0.269	0.937	0.644

Table 12: Tiny neural value-critic generalization audit. The value critic is trained on thresholds 1 and 3, then evaluated on held-out threshold 2. Exact tabular state lookup cannot match the evaluation threshold, so the TD signal comes from feature interpolation.

Estimator	r	MSE	Sign	Wait	Within
Group	0.364	0.0318	0.674	0.917	0.000
Neural TD	0.842	0.0104	0.875	0.418	0.038
Eval keys unseen in tabular train: 1.000; neural minus group r : 0.479.					

Tiny neural generalization. Table 12 is not an LLM-scale experiment, but it does remove the most brittle tabular assumption. On the held-out threshold, the group-relative broadcast reaches Pearson $r = 0.364$ against A^{π_b} token credit, while neural critic TD reaches $r = 0.842$ with lower calibrated MSE and lower delay-action magnitude. This strengthens the mechanism claim, while still leaving a larger transformer or nanoGPT-scale benchmark as future work.

9 Mechanism Analysis

Horizon length. The horizon cases show the central intuition. As trajectories lengthen and wait tokens accumulate, a response-level scalar is asked to explain increasingly heterogeneous token roles. The group-relative estimator can still rank complete rollouts, but that does not mean it assigns good token credit. The critic-style estimator retains access to progress state and remaining horizon, so it can separate helpful actions from delay actions even when the final reward is delayed.

Group size. Increasing group size improves the quality of group statistics only when the additional samples create meaningful reward dispersion. In the long-wait cases, larger groups help the group-relative estimator somewhat, but the broadcast problem remains: every token in a completion inherits the same normalized outcome. In settings where verifier outcomes are sparse or clustered, larger groups can spend many tokens for little additional advantage information.

Critic budget and coverage. The critic-budget axis is the strongest warning against a simplistic “critics are better” claim. With only two training groups, the critic has an exact-state hit rate below one half and the confidence interval crosses zero. With more coverage, the critic delta grows sharply. In a neural setting, the analogous quantity is not table hit rate but value-model calibration under the current policy distribution. A poorly covered critic should be treated as a risk, not a reason for confidence.

Observability. The observability cases show that a critic can still extract some value from coarse state, but a blind critic loses much of the temporal structure. The toy blind critic is intentionally simple; real long-horizon agents can become effectively blind for subtler reasons, including lossy compaction, missing tool state, stale memory, hidden environment variables, or value-model inputs that do not include the causal feature needed for future return.

Counterexample. The counterexample is the scientific control. In the blind-undercovered case, terminal group outcomes carry more usable information than a weak value table. The resulting group win is small but clear under the confidence-interval rule.

10 Estimator Regimes and Selection Criteria

The answer is conditional, and the condition is about information and reliability. Critic methods are more plausible when future return is predictable from prefix state, rollouts contain mixed token roles, and held-out value calibration is good enough to justify the extra model. Group-relative methods are more plausible when rewards are cheap, verifiable, and meaningfully dispersed within prompt groups, or when the critic would be blind, stale, or too expensive. Prefix-verifier and structural methods are plausible middle ground when they can create step-varying evidence without fitting a full critic. VIMPO-like policy-implied methods are plausible when the reference policy and KL-regularized value identity are reliable enough for the current training stage.

The practical selection rule is therefore not “use a critic” or “avoid a critic.” It is to estimate the information ceiling, the reliability error, and the resource vector. Report generated tokens, verifier calls, reference-policy scoring, critic memory, sampled branches, dense-logit cost, synchronization stalls, and wall-clock time beside reward changes.

11 Failure Modes and Testable Predictions

PPO/critic failures. Critics can overfit reward artifacts, become stale after policy drift, collapse on long sparse trajectories, or assign confident but wrong temporal credit.

GRPO failures. GRPO can suffer zero-variance groups, low-dispersion amplification, response length bias, scalar reward broadcast over harmful tokens, and group composition confounding prompt difficulty with policy quality [17, 20, 23, 30, 31, 37].

Dense distillation failures. Teacher/student thinking patterns may be incompatible; the teacher may offer no new information; on-policy rollouts may inflate in length and truncate; and privileged self-teacher traces may leak solution structure [13, 18, 19, 36].

Predictions. We predict that increasing horizon length and delay padding will reduce the Pearson alignment between response-level group advantages and exact or process labels. We predict that increasing group size will help GRPO mainly when it increases within-prompt reward dispersion. We predict that adding process rewards will narrow the PPO/GRPO credit gap, and that degrading critic observability will produce regimes where GRPO wins. Sampled-value methods should approach critic-like credit when the branching budget is high enough, but their inference cost should dominate in long agentic traces unless branch reuse or state sharing is available.

12 Threats to Validity

Toy dynamics. The toy has a small finite state and a known exact target. Most audits use tabular estimators, and the neural audit uses only a tiny value network on held-out toy thresholds. This is deliberately unlike frontier post-training. The benefit is interpretability; the cost is that transformer-scale generalization, distribution shift, optimizer dynamics, and reward-model error are not measured.

Estimator versus optimizer. The experiment compares advantage estimators, not complete PPO and GRPO training loops. A full optimizer could amplify or dampen estimator differences through clipping, KL schedules, entropy, minibatching, and policy parameterization. This paper therefore supports a mechanism hypothesis, not a leaderboard conclusion.

Simplified GRPO baseline. The group-relative baseline is intentionally simple. Future comparisons should include leave-one-out baselines, length-normalized objectives, Dr. GRPO-style corrections, DAPO-style dynamic sampling, token-level process rewards, and low-dispersion safeguards. A stronger GRPO family member may narrow the gap in some regimes.

Simplified critic. The main critic is tabular and trained from returns-to-go. The tiny neural audit checks interpolation across one held-out threshold, but it is still far from a large value model trained on language traces. Neural value models can generalize across states, but they can also hallucinate value structure. The toy’s exact-state hit rate is a proxy for coverage, not a complete model of critic reliability.

13 Reproducibility

The core experiment code has no external Python dependencies. The canonical checks are:

```
python3 -m unittest discover -s tests
./scripts/run_smoke.sh
python3 -m experiments.deep_matrix \
  --output-json results/deep_matrix_20seed.json \
  --output-csv results/deep_matrix_20seed.csv \
  --output-md results/deep_matrix_20seed.md \
  --figures-dir results/figures
python3 -m experiments.variance_credit_grid \
  --output-json results/variance_credit_grid_seed17.json \
  --output-md results/variance_credit_grid_seed17.md
python3 -m experiments.length_imbalance_audit \
  --output-json results/length_imbalance_audit_seedset.json \
  --output-md results/length_imbalance_audit_seedset.md
python3 -m experiments.token_cost_sensitivity \
  --output-json results/token_cost_sensitivity_20seed.json \
  --output-md results/token_cost_sensitivity_20seed.md
python3 -m experiments.closed_loop_credit_training \
  --output-json results/closed_loop_credit_training_10seed.json \
  --output-md results/closed_loop_credit_training_10seed.md
python3 -m experiments.neural_credit_generalization \
```

```

--output-json results/neural_credit_generalization_seedset.json \
--output-md results/neural_credit_generalization_seedset.md
python3 -m experiments.credit_phase_diagram \
--output-json results/credit_phase_diagram_seedset.json \
--output-md results/credit_phase_diagram_seedset.md
python3 -m experiments.policy_gradient_fidelity \
--output-json results/policy_gradient_fidelity_seed13.json \
--output-md results/policy_gradient_fidelity_seed13.md
./scripts/build_latex_paper.sh

```

The LaTeX paper is built with Tectonic when available and emitted as the full paper PDF artifact under `public/`.

14 Limitations

The toy environment has a finite known state and a tabular critic. Real post-training adds neural approximation, KL controllers, reward-model errors, tool failures, distributed rollout systems, stale policies, optimizer details, and data contamination risks. The GRPO baseline is intentionally simple; the next benchmark should include leave-one-out baselines, length-normalized variants, DAPO/Dr. GRPO-style corrections, process rewards, and dynamic sampling before making stronger claims.

15 Next Research Steps

nanoGPT scale. The next controlled step is a tiny sequence-model policy where the true environment is still known but both policy and value model must generalize over token histories. The held-out-threshold neural audit is a first approximation; a nanoGPT-scale benchmark is the next test of whether the mechanism survives transformer sequence modeling.

Agentic trace audits. For agentic research workflows, the key artifact is a trace schema that records tool state, compaction boundaries, blocked actions, reward calls, and process labels. Without that schema, value modeling and group-relative normalization cannot be audited fairly.

Cost-normalized benchmarks. Benchmark integrations should report cost-normalized improvement, not only final score. A critic may improve credit assignment but violate a deployment budget; a group method may be cheaper but lose sample efficiency or wall-clock time through synchronization stalls.

16 Conclusion

For long heterogeneous trajectories, critic-based PPO is a strong candidate when value targets are learnable and adequately covered. GRPO remains attractive when group reward contrast is reliable and critic modeling is too costly or poorly observed. VIMPO-like policy-implied signals, BRPO-like prefix baselines, structural anchors, sampled values, and process rewards all occupy meaningful middle ground. The paper’s central claim is therefore not “PPO always beats GRPO,” but a set of falsifiable boundaries for when token- or state-varying credit is worth its reliability and compute cost.

A Formal Objective Notes

Broadcasted group advantages. In a response-level group-relative update, a completion-level reward statistic is often applied to every trainable token in that completion. Let $g_i = (R_i - \mu_G)/(\sigma_G + \epsilon)$ be the group-normalized scalar for completion i . The token update is then proportional to $g_i \nabla_{\theta} \log \pi_{\theta}(a_{i,t} | s_{i,t})$ for many t . This is efficient, but in a single sampled completion it reinforces all sampled actions with the same scalar. That sample-level localization is weak whenever success depends on a small subset of steps and the rest are helpful delay, harmful detours, or irrelevant formatting. This does not contradict the unbiasedness of the total REINFORCE gradient; it states a token-fidelity limitation of trajectory-constant coefficients.

Temporal-difference advantages. A critic-style estimator replaces one response-level scalar with a local difference between predicted future return before and after an action. In the toy, this is a tabular one-step TD estimate. In language-model agents, the same idea may be implemented with value heads, separate value models, process reward models, or learned subgoal predictors. The central requirement is that the state representation contains enough information to make future return predictable.

Distillation losses. The dense distillation losses in the main text are written as KL objectives because the teacher signal is a distribution over next tokens. Other implementations may use sequence-level filtering, hidden-state matching, preference-filtered teacher traces, or mixed supervised/RL objectives. These variants can be useful without changing the taxonomy: they are dense imitation or self-imitation signals rather than direct policy-gradient reward optimization.

B Toy Environment Details

State. The state contains prompt identity, hidden progress, remaining horizon, and the features visible to the critic variant. The full critic observes enough to distinguish progress states. The coarse critic sees compressed state. The blind critic intentionally loses the feature needed for reliable future-return prediction.

Actions. The action set is deliberately small. **help** increases progress, **harm** decreases it, and **wait** consumes decision horizon without changing progress. This small action space makes exact dynamic programming feasible and makes the meaning of credit assignment visible. A language-model trace has richer tokens, but the policy-relative categories are recognizable: helpful substeps, harmful detours, delay actions, and formatting or null-like steps that future work should model separately.

Reward. The terminal verifier is sparse and thresholded. This makes the group-relative estimator plausible, because rollouts can be ranked by final success, but also creates the credit problem: the reward is delayed until after many actions.

Behavior-policy advantage. The behavior-policy advantage uses exact values under known dynamics. This is the scientific anchor of the paper. Without it, a method comparison could only say which estimator produced higher downstream score under one optimizer. With it, we can ask whether an estimator points in the same direction as the true policy-relative action credit.

C Case Design

The 18 fixed cases are grouped by axis. Horizon cases increase trace length and delay-action exposure. Group-size cases vary the number of completions used for within-prompt normalization. Critic-budget cases vary how much data the value table sees before evaluation. Observability cases degrade the state available to the critic. Sparse-reward cases make terminal success harder. The counterexample combines blindness and low coverage.

Table 13: Axis-level summary generated from the canonical matrix.

Axis	Cases	Group r	Critic r	Delta	Clear critic	Near tie	Clear group
counterexample	1	0.510	0.455	-0.055	0	0	1
critic_budget	4	0.352	0.617	0.265	3	1	0
group_size	4	0.305	0.868	0.563	4	0	0
horizon	4	0.379	0.908	0.530	4	0	0
observability	3	0.353	0.695	0.342	3	0	0
sparse_reward	2	0.296	0.878	0.582	2	0	0

D Full Case Summary

Table 14 gives the full 18-case summary. It is generated from the canonical matrix JSON rather than hand copied.

Table 14: Full 20-seed case summary. Delta is critic r minus group r .

Case	Axis	Read	Group r	Critic r	Δr	95% CI	Hit	Wait
horizon 04 baseline	horizon	critic	0.495	0.977	0.482	± 0.011	0.999	0.303
horizon 08 baseline	horizon	critic	0.390	0.925	0.535	± 0.015	0.996	0.365
horizon 12 baseline	horizon	critic	0.335	0.866	0.531	± 0.012	0.994	0.411
horizon 16 long wait	horizon	critic	0.293	0.865	0.571	± 0.013	0.996	0.539
group 02 long wait	group	critic	0.213	0.771	0.557	± 0.022	0.989	0.505
group 04 long wait	group	critic	0.301	0.851	0.550	± 0.014	0.994	0.503
group 08 long wait	group	critic	0.346	0.910	0.564	± 0.011	0.997	0.503
group 12 long wait	group	critic	0.358	0.940	0.582	± 0.007	0.998	0.501
budget 002 full	budget	near tie	0.352	0.367	0.015	± 0.027	0.476	0.388
budget 008 full	budget	critic	0.352	0.459	0.107	± 0.027	0.884	0.388
budget 032 full	budget	critic	0.352	0.735	0.383	± 0.021	0.979	0.388
budget 128 full	budget	critic	0.352	0.906	0.554	± 0.017	0.996	0.388
obs full	obs	critic	0.352	0.864	0.513	± 0.020	0.992	0.388
obs coarse	obs	critic	0.356	0.738	0.382	± 0.015	0.999	0.450
obs blind	obs	critic	0.353	0.482	0.129	± 0.017	1.000	0.444

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Table 14: Full 20-seed case summary, continued.

Case	Axis	Read	Group r	Critic r	Δr	95% CI	Hit	Wait
blind undercov.	ctr.	group	0.510	0.455	-0.055	± 0.015	0.693	0.329
sparse group04	sparse	critic	0.283	0.846	0.563	± 0.014	0.994	0.417
sparse group08	sparse	critic	0.310	0.910	0.600	± 0.008	0.997	0.420

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E Raw Seed-Level Rows

Table 15 records every case-by-seed row used to compute the paper’s aggregate claims. Publishing these rows makes the mean-win and confidence-interval interpretation auditable. The table is long by design: there are 18 cases and 20 seeds, for 360 rows.

Table 15: Raw case-by-seed estimator-fidelity rows from the deep matrix.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
horizon 04 baseline	11	0.514	0.974	0.460	0.998	0.422	0.290
horizon 04 baseline	29	0.480	0.976	0.495	0.998	0.391	0.312
horizon 04 baseline	47	0.477	0.982	0.505	1.000	0.396	0.286
horizon 04 baseline	83	0.522	0.975	0.452	1.000	0.458	0.331
horizon 04 baseline	131	0.471	0.971	0.500	1.000	0.391	0.297
horizon 04 baseline	173	0.505	0.971	0.466	1.000	0.422	0.296
horizon 04 baseline	211	0.480	0.986	0.506	1.000	0.422	0.310
horizon 04 baseline	257	0.503	0.980	0.477	1.000	0.427	0.305
horizon 04 baseline	307	0.493	0.985	0.492	1.000	0.438	0.317
horizon 04 baseline	359	0.456	0.981	0.525	1.000	0.422	0.295
horizon 04 baseline	401	0.526	0.981	0.456	1.000	0.438	0.298
horizon 04 baseline	463	0.491	0.981	0.490	1.000	0.417	0.309
horizon 04 baseline	509	0.494	0.983	0.489	1.000	0.385	0.294
horizon 04 baseline	571	0.463	0.969	0.506	0.997	0.396	0.330
horizon 04 baseline	631	0.471	0.977	0.506	0.997	0.474	0.265
horizon 04 baseline	701	0.499	0.973	0.475	1.000	0.448	0.304
horizon 04 baseline	761	0.492	0.945	0.453	0.998	0.443	0.299
horizon 04 baseline	823	0.548	0.974	0.427	1.000	0.464	0.312
horizon 04 baseline	887	0.510	0.985	0.475	1.000	0.422	0.301
horizon 04 baseline	947	0.508	0.985	0.477	1.000	0.448	0.304
horizon 08 baseline	11	0.410	0.898	0.488	0.995	0.542	0.361
horizon 08 baseline	29	0.370	0.917	0.548	0.998	0.531	0.382
horizon 08 baseline	47	0.391	0.900	0.509	0.996	0.542	0.395
horizon 08 baseline	83	0.365	0.923	0.558	0.996	0.490	0.369
horizon 08 baseline	131	0.404	0.890	0.486	0.992	0.552	0.362
horizon 08 baseline	173	0.395	0.924	0.529	0.999	0.542	0.373
horizon 08 baseline	211	0.402	0.919	0.517	0.994	0.547	0.347
horizon 08 baseline	257	0.401	0.928	0.527	0.997	0.552	0.373
horizon 08 baseline	307	0.386	0.945	0.559	1.000	0.583	0.364
horizon 08 baseline	359	0.339	0.957	0.618	1.000	0.526	0.376
horizon 08 baseline	401	0.410	0.900	0.491	0.986	0.516	0.334
horizon 08 baseline	463	0.374	0.925	0.551	0.998	0.589	0.380
horizon 08 baseline	509	0.390	0.919	0.529	0.998	0.599	0.369
horizon 08 baseline	571	0.376	0.973	0.596	0.999	0.568	0.347
horizon 08 baseline	631	0.379	0.947	0.568	1.000	0.510	0.341
horizon 08 baseline	701	0.400	0.925	0.526	0.996	0.583	0.404
horizon 08 baseline	761	0.380	0.916	0.536	0.996	0.547	0.350
horizon 08 baseline	823	0.420	0.913	0.494	0.990	0.578	0.357
horizon 08 baseline	887	0.408	0.937	0.530	0.995	0.604	0.364
horizon 08 baseline	947	0.405	0.945	0.540	1.000	0.500	0.358
horizon 12 baseline	11	0.314	0.868	0.554	0.999	0.609	0.415
horizon 12 baseline	29	0.348	0.919	0.571	0.995	0.589	0.400

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
horizon 12 baseline	47	0.321	0.870	0.549	0.997	0.677	0.417
horizon 12 baseline	83	0.346	0.869	0.524	0.994	0.573	0.396
horizon 12 baseline	131	0.339	0.878	0.540	0.994	0.604	0.397
horizon 12 baseline	173	0.357	0.836	0.479	0.994	0.630	0.416
horizon 12 baseline	211	0.337	0.860	0.523	0.996	0.667	0.390
horizon 12 baseline	257	0.346	0.893	0.547	0.997	0.656	0.437
horizon 12 baseline	307	0.324	0.877	0.554	0.992	0.667	0.412
horizon 12 baseline	359	0.349	0.846	0.497	0.995	0.599	0.397
horizon 12 baseline	401	0.349	0.870	0.521	0.993	0.630	0.412
horizon 12 baseline	463	0.348	0.842	0.494	0.997	0.656	0.416
horizon 12 baseline	509	0.330	0.887	0.557	0.992	0.661	0.425
horizon 12 baseline	571	0.326	0.840	0.514	0.986	0.661	0.424
horizon 12 baseline	631	0.342	0.847	0.505	0.990	0.604	0.392
horizon 12 baseline	701	0.316	0.895	0.578	0.996	0.651	0.445
horizon 12 baseline	761	0.349	0.884	0.536	0.990	0.625	0.405
horizon 12 baseline	823	0.341	0.847	0.507	0.990	0.573	0.399
horizon 12 baseline	887	0.314	0.852	0.538	0.998	0.646	0.402
horizon 12 baseline	947	0.314	0.844	0.530	0.993	0.620	0.421
horizon 16 long wait	11	0.302	0.862	0.560	0.998	0.667	0.513
horizon 16 long wait	29	0.303	0.884	0.581	0.998	0.667	0.536
horizon 16 long wait	47	0.266	0.866	0.601	0.998	0.718	0.538
horizon 16 long wait	83	0.319	0.885	0.565	0.996	0.620	0.529
horizon 16 long wait	131	0.292	0.871	0.579	0.995	0.671	0.542
horizon 16 long wait	173	0.297	0.869	0.572	0.997	0.606	0.552
horizon 16 long wait	211	0.292	0.875	0.583	0.996	0.718	0.546
horizon 16 long wait	257	0.299	0.865	0.566	0.991	0.694	0.532
horizon 16 long wait	307	0.264	0.909	0.645	0.998	0.671	0.539
horizon 16 long wait	359	0.287	0.847	0.560	0.994	0.741	0.546
horizon 16 long wait	401	0.291	0.809	0.517	0.993	0.718	0.539
horizon 16 long wait	463	0.299	0.860	0.561	0.997	0.662	0.552
horizon 16 long wait	509	0.306	0.875	0.568	0.995	0.662	0.545
horizon 16 long wait	571	0.297	0.826	0.529	0.994	0.657	0.547
horizon 16 long wait	631	0.305	0.840	0.534	0.995	0.667	0.517
horizon 16 long wait	701	0.291	0.845	0.554	0.996	0.713	0.527
horizon 16 long wait	761	0.316	0.878	0.562	0.997	0.736	0.562
horizon 16 long wait	823	0.287	0.866	0.579	0.995	0.657	0.532
horizon 16 long wait	887	0.272	0.895	0.623	0.998	0.713	0.539
horizon 16 long wait	947	0.279	0.868	0.589	0.998	0.694	0.545
group 02 long wait	11	0.174	0.665	0.492	0.968	0.641	0.507
group 02 long wait	29	0.227	0.774	0.547	0.991	0.719	0.496
group 02 long wait	47	0.223	0.792	0.569	0.994	0.656	0.495
group 02 long wait	83	0.245	0.709	0.464	0.987	0.719	0.487
group 02 long wait	131	0.189	0.758	0.569	0.998	0.625	0.514
group 02 long wait	173	0.212	0.794	0.582	0.995	0.734	0.499
group 02 long wait	211	0.216	0.773	0.557	0.987	0.578	0.489
group 02 long wait	257	0.236	0.775	0.539	0.986	0.641	0.515
group 02 long wait	307	0.187	0.811	0.624	0.995	0.672	0.533
group 02 long wait	359	0.212	0.699	0.487	0.980	0.625	0.453
group 02 long wait	401	0.195	0.720	0.525	0.996	0.672	0.528
group 02 long wait	463	0.184	0.805	0.621	0.993	0.609	0.496
group 02 long wait	509	0.223	0.754	0.532	0.979	0.656	0.515
group 02 long wait	571	0.247	0.878	0.631	0.986	0.656	0.536
group 02 long wait	631	0.214	0.777	0.563	0.998	0.641	0.492
group 02 long wait	701	0.263	0.748	0.485	0.978	0.609	0.488
group 02 long wait	761	0.225	0.783	0.558	0.998	0.672	0.490
group 02 long wait	823	0.207	0.795	0.588	0.994	0.688	0.498
group 02 long wait	887	0.203	0.774	0.571	0.987	0.734	0.537
group 02 long wait	947	0.187	0.827	0.641	0.991	0.703	0.542
group 04 long wait	11	0.273	0.788	0.515	0.991	0.680	0.488
group 04 long wait	29	0.305	0.861	0.557	0.998	0.734	0.495
group 04 long wait	47	0.265	0.794	0.529	0.988	0.656	0.481
group 04 long wait	83	0.300	0.874	0.574	1.000	0.664	0.503

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
group 04 long wait	131	0.323	0.867	0.544	0.998	0.625	0.524
group 04 long wait	173	0.333	0.796	0.464	0.985	0.656	0.465
group 04 long wait	211	0.291	0.846	0.555	0.999	0.609	0.507
group 04 long wait	257	0.308	0.875	0.567	0.996	0.664	0.533
group 04 long wait	307	0.331	0.879	0.548	0.991	0.641	0.495
group 04 long wait	359	0.289	0.859	0.570	0.993	0.609	0.471
group 04 long wait	401	0.298	0.909	0.611	0.999	0.656	0.544
group 04 long wait	463	0.293	0.880	0.587	0.995	0.570	0.487
group 04 long wait	509	0.300	0.825	0.525	0.992	0.617	0.516
group 04 long wait	571	0.321	0.879	0.558	0.988	0.688	0.514
group 04 long wait	631	0.303	0.841	0.538	0.998	0.680	0.485
group 04 long wait	701	0.291	0.876	0.585	0.997	0.688	0.517
group 04 long wait	761	0.296	0.871	0.575	1.000	0.688	0.491
group 04 long wait	823	0.296	0.857	0.561	0.982	0.664	0.508
group 04 long wait	887	0.282	0.799	0.517	0.995	0.672	0.508
group 04 long wait	947	0.330	0.847	0.517	0.997	0.633	0.516
group 08 long wait	11	0.337	0.903	0.566	0.997	0.656	0.506
group 08 long wait	29	0.345	0.915	0.569	0.998	0.652	0.493
group 08 long wait	47	0.301	0.907	0.607	0.998	0.680	0.506
group 08 long wait	83	0.356	0.931	0.575	0.997	0.625	0.460
group 08 long wait	131	0.373	0.931	0.558	0.996	0.637	0.522
group 08 long wait	173	0.353	0.865	0.512	0.996	0.633	0.497
group 08 long wait	211	0.357	0.927	0.570	0.998	0.621	0.503
group 08 long wait	257	0.344	0.938	0.595	0.998	0.617	0.505
group 08 long wait	307	0.333	0.925	0.592	0.998	0.664	0.511
group 08 long wait	359	0.350	0.891	0.541	0.995	0.660	0.505
group 08 long wait	401	0.343	0.924	0.581	0.998	0.660	0.515
group 08 long wait	463	0.340	0.900	0.560	0.999	0.609	0.518
group 08 long wait	509	0.333	0.925	0.592	0.996	0.691	0.503
group 08 long wait	571	0.333	0.867	0.534	0.994	0.672	0.526
group 08 long wait	631	0.366	0.904	0.538	0.997	0.609	0.517
group 08 long wait	701	0.342	0.899	0.556	0.998	0.668	0.505
group 08 long wait	761	0.343	0.945	0.602	0.998	0.586	0.494
group 08 long wait	823	0.346	0.887	0.541	0.989	0.617	0.484
group 08 long wait	887	0.369	0.914	0.545	0.997	0.613	0.484
group 08 long wait	947	0.350	0.900	0.550	0.996	0.660	0.513
group 12 long wait	11	0.365	0.924	0.559	0.999	0.646	0.496
group 12 long wait	29	0.367	0.935	0.568	0.999	0.633	0.491
group 12 long wait	47	0.338	0.958	0.620	0.999	0.659	0.511
group 12 long wait	83	0.372	0.939	0.568	0.999	0.674	0.498
group 12 long wait	131	0.353	0.958	0.605	1.000	0.638	0.531
group 12 long wait	173	0.355	0.948	0.593	0.998	0.690	0.490
group 12 long wait	211	0.351	0.948	0.596	0.998	0.617	0.488
group 12 long wait	257	0.352	0.940	0.588	0.999	0.607	0.511
group 12 long wait	307	0.367	0.943	0.576	1.000	0.654	0.502
group 12 long wait	359	0.348	0.938	0.590	0.998	0.667	0.493
group 12 long wait	401	0.363	0.937	0.574	0.997	0.674	0.502
group 12 long wait	463	0.356	0.919	0.564	0.997	0.646	0.520
group 12 long wait	509	0.352	0.942	0.590	0.996	0.635	0.490
group 12 long wait	571	0.368	0.928	0.560	0.996	0.628	0.511
group 12 long wait	631	0.371	0.937	0.566	0.998	0.656	0.507
group 12 long wait	701	0.367	0.945	0.578	1.000	0.693	0.513
group 12 long wait	761	0.344	0.934	0.590	0.998	0.659	0.500
group 12 long wait	823	0.348	0.947	0.599	0.997	0.617	0.486
group 12 long wait	887	0.356	0.942	0.585	0.998	0.635	0.496
group 12 long wait	947	0.363	0.942	0.579	0.999	0.656	0.494
budget 002 full	11	0.358	0.415	0.057	0.467	0.573	0.392
budget 002 full	29	0.395	0.405	0.009	0.476	0.620	0.400
budget 002 full	47	0.342	0.330	-0.012	0.536	0.609	0.385
budget 002 full	83	0.308	0.251	-0.056	0.439	0.630	0.409
budget 002 full	131	0.319	0.340	0.021	0.369	0.646	0.381
budget 002 full	173	0.328	0.441	0.113	0.514	0.562	0.381

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
budget 002 full	211	0.363	0.299	-0.064	0.501	0.630	0.398
budget 002 full	257	0.359	0.451	0.091	0.459	0.516	0.375
budget 002 full	307	0.357	0.267	-0.090	0.522	0.594	0.380
budget 002 full	359	0.313	0.414	0.101	0.376	0.568	0.364
budget 002 full	401	0.385	0.304	-0.081	0.506	0.573	0.374
budget 002 full	463	0.381	0.484	0.102	0.454	0.557	0.407
budget 002 full	509	0.329	0.358	0.029	0.489	0.604	0.381
budget 002 full	571	0.381	0.379	-0.003	0.432	0.599	0.397
budget 002 full	631	0.364	0.306	-0.058	0.470	0.562	0.388
budget 002 full	701	0.318	0.368	0.050	0.543	0.594	0.406
budget 002 full	761	0.329	0.375	0.046	0.503	0.651	0.413
budget 002 full	823	0.387	0.409	0.022	0.389	0.615	0.392
budget 002 full	887	0.377	0.368	-0.009	0.528	0.609	0.360
budget 002 full	947	0.342	0.383	0.040	0.539	0.599	0.372
budget 008 full	11	0.358	0.471	0.114	0.868	0.573	0.392
budget 008 full	29	0.395	0.536	0.141	0.873	0.620	0.400
budget 008 full	47	0.342	0.507	0.165	0.913	0.609	0.385
budget 008 full	83	0.308	0.470	0.163	0.856	0.630	0.409
budget 008 full	131	0.319	0.395	0.076	0.878	0.646	0.381
budget 008 full	173	0.328	0.567	0.239	0.885	0.562	0.381
budget 008 full	211	0.363	0.466	0.104	0.846	0.630	0.398
budget 008 full	257	0.359	0.517	0.158	0.917	0.516	0.375
budget 008 full	307	0.357	0.399	0.042	0.926	0.594	0.380
budget 008 full	359	0.313	0.530	0.217	0.872	0.568	0.364
budget 008 full	401	0.385	0.480	0.096	0.885	0.573	0.374
budget 008 full	463	0.381	0.455	0.074	0.958	0.557	0.407
budget 008 full	509	0.329	0.435	0.106	0.913	0.604	0.381
budget 008 full	571	0.381	0.465	0.084	0.875	0.599	0.397
budget 008 full	631	0.364	0.353	-0.012	0.828	0.562	0.388
budget 008 full	701	0.318	0.422	0.104	0.884	0.594	0.406
budget 008 full	761	0.329	0.348	0.020	0.847	0.651	0.413
budget 008 full	823	0.387	0.429	0.042	0.849	0.615	0.392
budget 008 full	887	0.377	0.501	0.124	0.903	0.609	0.360
budget 008 full	947	0.342	0.434	0.092	0.902	0.599	0.372
budget 032 full	11	0.358	0.715	0.357	0.975	0.573	0.392
budget 032 full	29	0.395	0.761	0.365	0.974	0.620	0.400
budget 032 full	47	0.342	0.749	0.407	0.987	0.609	0.385
budget 032 full	83	0.308	0.709	0.402	0.975	0.630	0.409
budget 032 full	131	0.319	0.715	0.396	0.974	0.646	0.381
budget 032 full	173	0.328	0.828	0.500	0.980	0.562	0.381
budget 032 full	211	0.363	0.667	0.305	0.971	0.630	0.398
budget 032 full	257	0.359	0.795	0.436	0.993	0.516	0.375
budget 032 full	307	0.357	0.708	0.351	0.983	0.594	0.380
budget 032 full	359	0.313	0.780	0.467	0.982	0.568	0.364
budget 032 full	401	0.385	0.691	0.306	0.973	0.573	0.374
budget 032 full	463	0.381	0.757	0.375	0.987	0.557	0.407
budget 032 full	509	0.329	0.672	0.343	0.981	0.604	0.381
budget 032 full	571	0.381	0.717	0.336	0.979	0.599	0.397
budget 032 full	631	0.364	0.720	0.356	0.961	0.562	0.388
budget 032 full	701	0.318	0.700	0.382	0.969	0.594	0.406
budget 032 full	761	0.329	0.700	0.372	0.973	0.651	0.413
budget 032 full	823	0.387	0.798	0.411	0.993	0.615	0.392
budget 032 full	887	0.377	0.771	0.394	0.985	0.609	0.360
budget 032 full	947	0.342	0.740	0.397	0.984	0.599	0.372
budget 128 full	11	0.358	0.907	0.549	0.994	0.573	0.392
budget 128 full	29	0.395	0.884	0.489	0.996	0.620	0.400
budget 128 full	47	0.342	0.909	0.567	0.997	0.609	0.385
budget 128 full	83	0.308	0.896	0.588	0.996	0.630	0.409
budget 128 full	131	0.319	0.899	0.580	0.992	0.646	0.381
budget 128 full	173	0.328	0.942	0.614	1.000	0.562	0.381
budget 128 full	211	0.363	0.873	0.510	0.993	0.630	0.398
budget 128 full	257	0.359	0.917	0.557	0.998	0.516	0.375

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
budget 128 full	307	0.357	0.880	0.523	0.993	0.594	0.380
budget 128 full	359	0.313	0.938	0.625	0.997	0.568	0.364
budget 128 full	401	0.385	0.905	0.521	0.992	0.573	0.374
budget 128 full	463	0.381	0.915	0.533	0.996	0.557	0.407
budget 128 full	509	0.329	0.923	0.594	0.993	0.604	0.381
budget 128 full	571	0.381	0.880	0.498	0.997	0.599	0.397
budget 128 full	631	0.364	0.881	0.516	0.987	0.562	0.388
budget 128 full	701	0.318	0.909	0.591	0.999	0.594	0.406
budget 128 full	761	0.329	0.897	0.568	0.997	0.651	0.413
budget 128 full	823	0.387	0.935	0.549	0.996	0.615	0.392
budget 128 full	887	0.377	0.914	0.537	0.998	0.609	0.360
budget 128 full	947	0.342	0.919	0.577	0.998	0.599	0.372
obs full	11	0.358	0.843	0.485	0.991	0.573	0.392
obs full	29	0.395	0.856	0.460	0.992	0.620	0.400
obs full	47	0.342	0.867	0.525	0.996	0.609	0.385
obs full	83	0.308	0.872	0.564	0.993	0.630	0.409
obs full	131	0.319	0.843	0.524	0.984	0.646	0.381
obs full	173	0.328	0.935	0.607	1.000	0.562	0.381
obs full	211	0.363	0.835	0.473	0.987	0.630	0.398
obs full	257	0.359	0.871	0.512	0.995	0.516	0.375
obs full	307	0.357	0.868	0.510	0.991	0.594	0.380
obs full	359	0.313	0.890	0.577	0.993	0.568	0.364
obs full	401	0.385	0.809	0.424	0.989	0.573	0.374
obs full	463	0.381	0.870	0.489	0.994	0.557	0.407
obs full	509	0.329	0.893	0.564	0.989	0.604	0.381
obs full	571	0.381	0.836	0.454	0.993	0.599	0.397
obs full	631	0.364	0.845	0.481	0.984	0.562	0.388
obs full	701	0.318	0.831	0.513	0.989	0.594	0.406
obs full	761	0.329	0.859	0.531	0.996	0.651	0.413
obs full	823	0.387	0.891	0.504	0.996	0.615	0.392
obs full	887	0.377	0.894	0.517	0.995	0.609	0.360
obs full	947	0.342	0.881	0.539	0.996	0.599	0.372
obs coarse	11	0.361	0.701	0.341	1.000	0.568	0.440
obs coarse	29	0.378	0.710	0.331	1.000	0.630	0.453
obs coarse	47	0.351	0.702	0.352	1.000	0.646	0.453
obs coarse	83	0.321	0.773	0.452	1.000	0.578	0.456
obs coarse	131	0.344	0.727	0.382	1.000	0.615	0.434
obs coarse	173	0.355	0.728	0.373	1.000	0.609	0.446
obs coarse	211	0.366	0.735	0.370	0.997	0.630	0.453
obs coarse	257	0.380	0.767	0.386	1.000	0.547	0.441
obs coarse	307	0.342	0.710	0.368	1.000	0.620	0.452
obs coarse	359	0.328	0.760	0.432	1.000	0.599	0.461
obs coarse	401	0.371	0.758	0.387	1.000	0.625	0.443
obs coarse	463	0.367	0.703	0.336	1.000	0.604	0.457
obs coarse	509	0.308	0.750	0.442	1.000	0.635	0.441
obs coarse	571	0.379	0.760	0.382	1.000	0.625	0.464
obs coarse	631	0.357	0.757	0.400	0.998	0.557	0.433
obs coarse	701	0.345	0.714	0.369	0.998	0.656	0.477
obs coarse	761	0.359	0.707	0.348	0.997	0.620	0.464
obs coarse	823	0.375	0.761	0.386	1.000	0.573	0.445
obs coarse	887	0.380	0.756	0.375	0.999	0.594	0.425
obs coarse	947	0.347	0.784	0.437	1.000	0.609	0.454
obs blind	11	0.358	0.400	0.042	1.000	0.568	0.442
obs blind	29	0.376	0.501	0.125	1.000	0.630	0.444
obs blind	47	0.342	0.423	0.081	1.000	0.651	0.454
obs blind	83	0.321	0.459	0.138	1.000	0.589	0.457
obs blind	131	0.340	0.463	0.124	1.000	0.615	0.426
obs blind	173	0.360	0.485	0.125	1.000	0.599	0.442
obs blind	211	0.355	0.519	0.164	1.000	0.615	0.450
obs blind	257	0.384	0.487	0.103	1.000	0.542	0.435
obs blind	307	0.336	0.417	0.081	1.000	0.615	0.441
obs blind	359	0.330	0.521	0.190	1.000	0.604	0.444

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
obs blind	401	0.360	0.456	0.096	1.000	0.620	0.437
obs blind	463	0.372	0.477	0.104	1.000	0.594	0.451
obs blind	509	0.297	0.486	0.189	1.000	0.641	0.436
obs blind	571	0.377	0.532	0.155	1.000	0.620	0.454
obs blind	631	0.349	0.491	0.143	1.000	0.562	0.426
obs blind	701	0.341	0.487	0.145	1.000	0.641	0.470
obs blind	761	0.352	0.486	0.134	1.000	0.630	0.468
obs blind	823	0.375	0.551	0.177	1.000	0.562	0.435
obs blind	887	0.374	0.500	0.126	1.000	0.604	0.423
obs blind	947	0.351	0.499	0.148	1.000	0.604	0.449
blind undercov.	11	0.518	0.460	-0.058	0.694	0.398	0.313
blind undercov.	29	0.472	0.496	0.024	0.697	0.414	0.324
blind undercov.	47	0.518	0.450	-0.068	0.683	0.434	0.330
blind undercov.	83	0.512	0.413	-0.099	0.689	0.430	0.357
blind undercov.	131	0.525	0.464	-0.061	0.681	0.434	0.315
blind undercov.	173	0.525	0.505	-0.020	0.680	0.469	0.316
blind undercov.	211	0.508	0.494	-0.015	0.670	0.461	0.327
blind undercov.	257	0.502	0.432	-0.070	0.699	0.406	0.309
blind undercov.	307	0.536	0.478	-0.057	0.697	0.477	0.339
blind undercov.	359	0.525	0.420	-0.105	0.706	0.426	0.315
blind undercov.	401	0.502	0.408	-0.094	0.704	0.398	0.334
blind undercov.	463	0.503	0.471	-0.031	0.692	0.391	0.352
blind undercov.	509	0.525	0.495	-0.030	0.705	0.434	0.348
blind undercov.	571	0.509	0.496	-0.014	0.692	0.418	0.362
blind undercov.	631	0.473	0.432	-0.040	0.683	0.418	0.297
blind undercov.	701	0.513	0.463	-0.050	0.686	0.430	0.323
blind undercov.	761	0.498	0.401	-0.098	0.698	0.438	0.316
blind undercov.	823	0.520	0.448	-0.072	0.703	0.449	0.332
blind undercov.	887	0.496	0.447	-0.049	0.706	0.426	0.343
blind undercov.	947	0.513	0.419	-0.094	0.688	0.449	0.322
sparse group04	11	0.282	0.832	0.551	0.994	0.486	0.407
sparse group04	29	0.290	0.848	0.558	0.997	0.528	0.442
sparse group04	47	0.301	0.829	0.528	0.991	0.472	0.413
sparse group04	83	0.246	0.853	0.607	0.998	0.458	0.432
sparse group04	131	0.320	0.838	0.518	0.996	0.479	0.433
sparse group04	173	0.309	0.880	0.571	0.994	0.472	0.399
sparse group04	211	0.286	0.852	0.566	0.993	0.493	0.404
sparse group04	257	0.274	0.858	0.584	0.996	0.431	0.419
sparse group04	307	0.303	0.857	0.555	0.991	0.403	0.412
sparse group04	359	0.244	0.820	0.575	0.992	0.493	0.412
sparse group04	401	0.278	0.817	0.539	0.988	0.479	0.417
sparse group04	463	0.305	0.870	0.564	0.996	0.507	0.435
sparse group04	509	0.296	0.864	0.568	0.986	0.424	0.439
sparse group04	571	0.251	0.836	0.586	0.996	0.472	0.408
sparse group04	631	0.287	0.845	0.558	0.993	0.451	0.399
sparse group04	701	0.262	0.835	0.573	0.998	0.472	0.435
sparse group04	761	0.286	0.781	0.495	0.987	0.493	0.429
sparse group04	823	0.259	0.908	0.649	0.996	0.500	0.410
sparse group04	887	0.287	0.861	0.574	0.994	0.479	0.412
sparse group04	947	0.289	0.829	0.540	0.998	0.500	0.385
sparse group08	11	0.337	0.924	0.587	0.999	0.497	0.409
sparse group08	29	0.296	0.897	0.601	1.000	0.455	0.434
sparse group08	47	0.324	0.894	0.570	0.995	0.462	0.425
sparse group08	83	0.302	0.915	0.613	0.996	0.455	0.417
sparse group08	131	0.307	0.931	0.624	0.998	0.458	0.430
sparse group08	173	0.312	0.911	0.599	0.995	0.469	0.415
sparse group08	211	0.299	0.921	0.622	0.999	0.455	0.405
sparse group08	257	0.318	0.901	0.582	0.999	0.462	0.414
sparse group08	307	0.302	0.912	0.610	0.998	0.417	0.419
sparse group08	359	0.324	0.894	0.570	0.995	0.472	0.419
sparse group08	401	0.296	0.885	0.589	0.996	0.479	0.417
sparse group08	463	0.302	0.931	0.629	0.999	0.507	0.429

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Table 15: Raw case-by-seed estimator-fidelity rows, continued.

Case	Seed	Group r	Critic r	Δr	Hit	Success	Wait
sparse group08	509	0.318	0.923	0.605	0.994	0.476	0.428
sparse group08	571	0.302	0.916	0.614	0.999	0.427	0.415
sparse group08	631	0.321	0.904	0.583	0.997	0.465	0.417
sparse group08	701	0.316	0.903	0.587	0.999	0.490	0.434
sparse group08	761	0.298	0.914	0.616	0.997	0.479	0.434
sparse group08	823	0.311	0.903	0.591	0.996	0.444	0.402
sparse group08	887	0.306	0.916	0.610	0.997	0.476	0.419
sparse group08	947	0.309	0.911	0.602	0.998	0.465	0.416

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F Raw Error and Dispersion Rows

Table 16 records the calibrated-MSE side of the same matrix. These rows help distinguish two different claims: a method can have higher correlation with behavior-policy credit while also changing calibrated error, zero-variance group frequency, or success-rate mix.

Table 16: Raw case-by-seed calibrated-error rows from the deep matrix.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
horizon 04 baseline	11	0.050	0.004	-0.046	0.000	0.290	0.422
horizon 04 baseline	29	0.053	0.003	-0.050	0.000	0.312	0.391
horizon 04 baseline	47	0.047	0.002	-0.045	0.000	0.286	0.396
horizon 04 baseline	83	0.044	0.003	-0.041	0.000	0.331	0.458
horizon 04 baseline	131	0.047	0.003	-0.044	0.000	0.297	0.391
horizon 04 baseline	173	0.049	0.004	-0.046	0.000	0.296	0.422
horizon 04 baseline	211	0.053	0.002	-0.051	0.000	0.310	0.422
horizon 04 baseline	257	0.054	0.003	-0.051	0.000	0.305	0.427
horizon 04 baseline	307	0.044	0.002	-0.042	0.000	0.317	0.438
horizon 04 baseline	359	0.058	0.003	-0.055	0.000	0.295	0.422
horizon 04 baseline	401	0.051	0.003	-0.048	0.000	0.298	0.438
horizon 04 baseline	463	0.051	0.003	-0.048	0.000	0.309	0.417
horizon 04 baseline	509	0.054	0.002	-0.052	0.000	0.294	0.385
horizon 04 baseline	571	0.052	0.004	-0.048	0.000	0.330	0.396
horizon 04 baseline	631	0.056	0.003	-0.053	0.000	0.265	0.474
horizon 04 baseline	701	0.052	0.004	-0.049	0.000	0.304	0.448
horizon 04 baseline	761	0.047	0.007	-0.040	0.000	0.299	0.443
horizon 04 baseline	823	0.045	0.003	-0.042	0.000	0.312	0.464
horizon 04 baseline	887	0.049	0.002	-0.047	0.000	0.301	0.422
horizon 04 baseline	947	0.050	0.002	-0.048	0.000	0.304	0.448
horizon 08 baseline	11	0.034	0.008	-0.026	0.000	0.361	0.542
horizon 08 baseline	29	0.036	0.007	-0.029	0.000	0.382	0.531
horizon 08 baseline	47	0.036	0.008	-0.028	0.000	0.395	0.542
horizon 08 baseline	83	0.036	0.006	-0.030	0.000	0.369	0.490
horizon 08 baseline	131	0.030	0.007	-0.022	0.000	0.362	0.552
horizon 08 baseline	173	0.038	0.007	-0.031	0.000	0.373	0.542
horizon 08 baseline	211	0.038	0.007	-0.031	0.000	0.347	0.547
horizon 08 baseline	257	0.035	0.006	-0.029	0.000	0.373	0.552
horizon 08 baseline	307	0.034	0.004	-0.030	0.000	0.364	0.583
horizon 08 baseline	359	0.039	0.004	-0.035	0.000	0.376	0.526
horizon 08 baseline	401	0.035	0.008	-0.027	0.000	0.334	0.516
horizon 08 baseline	463	0.037	0.006	-0.031	0.000	0.380	0.589
horizon 08 baseline	509	0.034	0.006	-0.028	0.000	0.369	0.599
horizon 08 baseline	571	0.040	0.003	-0.037	0.000	0.347	0.568
horizon 08 baseline	631	0.035	0.004	-0.031	0.000	0.341	0.510
horizon 08 baseline	701	0.033	0.006	-0.027	0.000	0.404	0.583
horizon 08 baseline	761	0.037	0.007	-0.030	0.000	0.350	0.547

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
horizon 08 baseline	823	0.031	0.006	-0.025	0.000	0.357	0.578
horizon 08 baseline	887	0.032	0.005	-0.028	0.000	0.364	0.604
horizon 08 baseline	947	0.037	0.005	-0.032	0.000	0.358	0.500
horizon 12 baseline	11	0.024	0.007	-0.017	0.000	0.415	0.609
horizon 12 baseline	29	0.028	0.005	-0.023	0.000	0.400	0.589
horizon 12 baseline	47	0.022	0.006	-0.016	0.000	0.417	0.677
horizon 12 baseline	83	0.027	0.007	-0.019	0.000	0.396	0.573
horizon 12 baseline	131	0.026	0.007	-0.019	0.000	0.397	0.604
horizon 12 baseline	173	0.026	0.009	-0.017	0.000	0.416	0.630
horizon 12 baseline	211	0.025	0.007	-0.018	0.000	0.390	0.667
horizon 12 baseline	257	0.023	0.005	-0.017	0.000	0.437	0.656
horizon 12 baseline	307	0.025	0.006	-0.018	0.000	0.412	0.667
horizon 12 baseline	359	0.024	0.008	-0.016	0.000	0.397	0.599
horizon 12 baseline	401	0.025	0.007	-0.018	0.000	0.412	0.630
horizon 12 baseline	463	0.022	0.007	-0.015	0.000	0.416	0.656
horizon 12 baseline	509	0.025	0.006	-0.019	0.000	0.425	0.661
horizon 12 baseline	571	0.024	0.008	-0.016	0.000	0.424	0.661
horizon 12 baseline	631	0.028	0.009	-0.019	0.000	0.392	0.604
horizon 12 baseline	701	0.024	0.005	-0.019	0.000	0.445	0.651
horizon 12 baseline	761	0.025	0.006	-0.019	0.000	0.405	0.625
horizon 12 baseline	823	0.027	0.009	-0.018	0.000	0.399	0.573
horizon 12 baseline	887	0.025	0.008	-0.017	0.000	0.402	0.646
horizon 12 baseline	947	0.025	0.008	-0.017	0.000	0.421	0.620
horizon 16 long wait	11	0.021	0.006	-0.015	0.000	0.513	0.667
horizon 16 long wait	29	0.018	0.004	-0.014	0.000	0.536	0.667
horizon 16 long wait	47	0.017	0.005	-0.013	0.000	0.538	0.718
horizon 16 long wait	83	0.020	0.005	-0.016	0.000	0.529	0.620
horizon 16 long wait	131	0.019	0.005	-0.014	0.000	0.542	0.671
horizon 16 long wait	173	0.019	0.005	-0.014	0.000	0.552	0.606
horizon 16 long wait	211	0.019	0.005	-0.014	0.000	0.546	0.718
horizon 16 long wait	257	0.017	0.005	-0.012	0.000	0.532	0.694
horizon 16 long wait	307	0.023	0.004	-0.019	0.000	0.539	0.671
horizon 16 long wait	359	0.016	0.005	-0.011	0.000	0.546	0.741
horizon 16 long wait	401	0.017	0.006	-0.011	0.000	0.539	0.718
horizon 16 long wait	463	0.018	0.005	-0.013	0.000	0.552	0.662
horizon 16 long wait	509	0.019	0.005	-0.014	0.000	0.545	0.662
horizon 16 long wait	571	0.018	0.006	-0.012	0.000	0.547	0.657
horizon 16 long wait	631	0.019	0.006	-0.013	0.000	0.517	0.667
horizon 16 long wait	701	0.017	0.005	-0.012	0.000	0.527	0.713
horizon 16 long wait	761	0.015	0.004	-0.011	0.000	0.562	0.736
horizon 16 long wait	823	0.022	0.006	-0.016	0.000	0.532	0.657
horizon 16 long wait	887	0.020	0.004	-0.016	0.000	0.539	0.713
horizon 16 long wait	947	0.018	0.005	-0.013	0.000	0.545	0.694
group 02 long wait	11	0.024	0.014	-0.010	0.031	0.507	0.641
group 02 long wait	29	0.024	0.010	-0.014	0.094	0.496	0.719
group 02 long wait	47	0.025	0.010	-0.015	0.062	0.495	0.656
group 02 long wait	83	0.022	0.011	-0.010	0.125	0.487	0.719
group 02 long wait	131	0.024	0.010	-0.013	0.094	0.514	0.625
group 02 long wait	173	0.024	0.009	-0.015	0.125	0.499	0.734
group 02 long wait	211	0.025	0.010	-0.014	0.031	0.489	0.578
group 02 long wait	257	0.024	0.010	-0.014	0.156	0.515	0.641
group 02 long wait	307	0.026	0.009	-0.017	0.062	0.533	0.672
group 02 long wait	359	0.026	0.014	-0.012	0.000	0.453	0.625
group 02 long wait	401	0.024	0.012	-0.012	0.062	0.528	0.672
group 02 long wait	463	0.022	0.008	-0.014	0.062	0.496	0.609
group 02 long wait	509	0.028	0.013	-0.015	0.125	0.515	0.656
group 02 long wait	571	0.035	0.009	-0.026	0.062	0.536	0.656
group 02 long wait	631	0.027	0.011	-0.016	0.000	0.492	0.641
group 02 long wait	701	0.028	0.013	-0.015	0.062	0.488	0.609
group 02 long wait	761	0.027	0.011	-0.016	0.031	0.490	0.672
group 02 long wait	823	0.024	0.009	-0.015	0.062	0.498	0.688
group 02 long wait	887	0.017	0.007	-0.010	0.031	0.537	0.734

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
group 02 long wait	947	0.024	0.008	-0.016	0.031	0.542	0.703
group 04 long wait	11	0.023	0.010	-0.014	0.000	0.488	0.680
group 04 long wait	29	0.022	0.006	-0.016	0.000	0.495	0.734
group 04 long wait	47	0.023	0.009	-0.014	0.000	0.481	0.656
group 04 long wait	83	0.022	0.006	-0.016	0.000	0.503	0.664
group 04 long wait	131	0.025	0.007	-0.018	0.000	0.524	0.625
group 04 long wait	173	0.025	0.010	-0.015	0.000	0.465	0.656
group 04 long wait	211	0.026	0.008	-0.018	0.000	0.507	0.609
group 04 long wait	257	0.021	0.006	-0.016	0.000	0.533	0.664
group 04 long wait	307	0.025	0.006	-0.019	0.000	0.495	0.641
group 04 long wait	359	0.029	0.008	-0.021	0.000	0.471	0.609
group 04 long wait	401	0.022	0.004	-0.018	0.000	0.544	0.656
group 04 long wait	463	0.027	0.007	-0.021	0.000	0.487	0.570
group 04 long wait	509	0.028	0.010	-0.018	0.000	0.516	0.617
group 04 long wait	571	0.026	0.007	-0.019	0.000	0.514	0.688
group 04 long wait	631	0.021	0.007	-0.014	0.000	0.485	0.680
group 04 long wait	701	0.023	0.006	-0.017	0.000	0.517	0.688
group 04 long wait	761	0.022	0.006	-0.017	0.000	0.491	0.688
group 04 long wait	823	0.024	0.007	-0.017	0.000	0.508	0.664
group 04 long wait	887	0.024	0.009	-0.015	0.000	0.508	0.672
group 04 long wait	947	0.020	0.007	-0.014	0.000	0.516	0.633
group 08 long wait	11	0.024	0.005	-0.019	0.000	0.506	0.656
group 08 long wait	29	0.026	0.005	-0.021	0.000	0.493	0.652
group 08 long wait	47	0.022	0.004	-0.018	0.000	0.506	0.680
group 08 long wait	83	0.025	0.004	-0.021	0.000	0.460	0.625
group 08 long wait	131	0.022	0.003	-0.019	0.000	0.522	0.637
group 08 long wait	173	0.025	0.007	-0.018	0.000	0.497	0.633
group 08 long wait	211	0.026	0.004	-0.021	0.000	0.503	0.621
group 08 long wait	257	0.025	0.003	-0.021	0.000	0.505	0.617
group 08 long wait	307	0.026	0.004	-0.021	0.000	0.511	0.664
group 08 long wait	359	0.024	0.006	-0.019	0.000	0.505	0.660
group 08 long wait	401	0.024	0.004	-0.020	0.000	0.515	0.660
group 08 long wait	463	0.025	0.005	-0.020	0.000	0.518	0.609
group 08 long wait	509	0.024	0.004	-0.020	0.000	0.503	0.691
group 08 long wait	571	0.023	0.007	-0.017	0.000	0.526	0.672
group 08 long wait	631	0.021	0.004	-0.017	0.000	0.517	0.609
group 08 long wait	701	0.021	0.005	-0.017	0.000	0.505	0.668
group 08 long wait	761	0.026	0.003	-0.023	0.000	0.494	0.586
group 08 long wait	823	0.024	0.006	-0.018	0.000	0.484	0.617
group 08 long wait	887	0.026	0.005	-0.021	0.000	0.484	0.613
group 08 long wait	947	0.020	0.004	-0.016	0.000	0.513	0.660
group 12 long wait	11	0.024	0.004	-0.020	0.000	0.496	0.646
group 12 long wait	29	0.026	0.004	-0.023	0.000	0.491	0.633
group 12 long wait	47	0.023	0.002	-0.021	0.000	0.511	0.659
group 12 long wait	83	0.023	0.003	-0.020	0.000	0.498	0.674
group 12 long wait	131	0.024	0.002	-0.022	0.000	0.531	0.638
group 12 long wait	173	0.023	0.003	-0.021	0.000	0.490	0.690
group 12 long wait	211	0.028	0.003	-0.024	0.000	0.488	0.617
group 12 long wait	257	0.025	0.003	-0.021	0.000	0.511	0.607
group 12 long wait	307	0.025	0.003	-0.022	0.000	0.502	0.654
group 12 long wait	359	0.024	0.003	-0.021	0.000	0.493	0.667
group 12 long wait	401	0.023	0.003	-0.020	0.000	0.502	0.674
group 12 long wait	463	0.024	0.004	-0.020	0.000	0.520	0.646
group 12 long wait	509	0.025	0.003	-0.022	0.000	0.490	0.635
group 12 long wait	571	0.024	0.004	-0.020	0.000	0.511	0.628
group 12 long wait	631	0.021	0.003	-0.018	0.000	0.507	0.656
group 12 long wait	701	0.021	0.003	-0.018	0.000	0.513	0.693
group 12 long wait	761	0.024	0.003	-0.020	0.000	0.500	0.659
group 12 long wait	823	0.026	0.003	-0.023	0.000	0.486	0.617
group 12 long wait	887	0.026	0.003	-0.023	0.000	0.496	0.635
group 12 long wait	947	0.023	0.003	-0.020	0.000	0.494	0.656
budget 002 full	11	0.029	0.027	-0.001	0.000	0.392	0.573

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
budget 002 full	29	0.028	0.028	-0.000	0.000	0.400	0.620
budget 002 full	47	0.029	0.029	0.000	0.000	0.385	0.609
budget 002 full	83	0.027	0.028	0.001	0.000	0.409	0.630
budget 002 full	131	0.028	0.028	-0.000	0.000	0.381	0.646
budget 002 full	173	0.036	0.033	-0.004	0.000	0.381	0.562
budget 002 full	211	0.028	0.029	0.001	0.000	0.398	0.630
budget 002 full	257	0.034	0.031	-0.003	0.000	0.375	0.516
budget 002 full	307	0.029	0.031	0.002	0.000	0.380	0.594
budget 002 full	359	0.035	0.032	-0.003	0.000	0.364	0.568
budget 002 full	401	0.030	0.032	0.002	0.000	0.374	0.573
budget 002 full	463	0.033	0.029	-0.003	0.000	0.407	0.557
budget 002 full	509	0.030	0.030	-0.001	0.000	0.381	0.604
budget 002 full	571	0.031	0.031	0.000	0.000	0.397	0.599
budget 002 full	631	0.031	0.032	0.001	0.000	0.388	0.562
budget 002 full	701	0.032	0.031	-0.001	0.000	0.406	0.594
budget 002 full	761	0.026	0.025	-0.001	0.000	0.413	0.651
budget 002 full	823	0.027	0.026	-0.001	0.000	0.392	0.615
budget 002 full	887	0.031	0.032	0.000	0.000	0.360	0.609
budget 002 full	947	0.033	0.032	-0.001	0.000	0.372	0.599
budget 008 full	11	0.029	0.026	-0.003	0.000	0.392	0.573
budget 008 full	29	0.028	0.024	-0.004	0.000	0.400	0.620
budget 008 full	47	0.029	0.024	-0.005	0.000	0.385	0.609
budget 008 full	83	0.027	0.024	-0.004	0.000	0.409	0.630
budget 008 full	131	0.028	0.026	-0.002	0.000	0.381	0.646
budget 008 full	173	0.036	0.027	-0.009	0.000	0.381	0.562
budget 008 full	211	0.028	0.025	-0.003	0.000	0.398	0.630
budget 008 full	257	0.034	0.028	-0.005	0.000	0.375	0.516
budget 008 full	307	0.029	0.028	-0.001	0.000	0.380	0.594
budget 008 full	359	0.035	0.028	-0.007	0.000	0.364	0.568
budget 008 full	401	0.030	0.027	-0.003	0.000	0.374	0.573
budget 008 full	463	0.033	0.030	-0.002	0.000	0.407	0.557
budget 008 full	509	0.030	0.028	-0.003	0.000	0.381	0.604
budget 008 full	571	0.031	0.028	-0.003	0.000	0.397	0.599
budget 008 full	631	0.031	0.031	0.000	0.000	0.388	0.562
budget 008 full	701	0.032	0.029	-0.003	0.000	0.406	0.594
budget 008 full	761	0.026	0.026	-0.000	0.000	0.413	0.651
budget 008 full	823	0.027	0.026	-0.001	0.000	0.392	0.615
budget 008 full	887	0.031	0.027	-0.004	0.000	0.360	0.609
budget 008 full	947	0.033	0.030	-0.003	0.000	0.372	0.599
budget 032 full	11	0.029	0.016	-0.013	0.000	0.392	0.573
budget 032 full	29	0.028	0.014	-0.014	0.000	0.400	0.620
budget 032 full	47	0.029	0.014	-0.014	0.000	0.385	0.609
budget 032 full	83	0.027	0.015	-0.012	0.000	0.409	0.630
budget 032 full	131	0.028	0.015	-0.013	0.000	0.381	0.646
budget 032 full	173	0.036	0.013	-0.023	0.000	0.381	0.562
budget 032 full	211	0.028	0.018	-0.010	0.000	0.398	0.630
budget 032 full	257	0.034	0.014	-0.019	0.000	0.375	0.516
budget 032 full	307	0.029	0.017	-0.013	0.000	0.380	0.594
budget 032 full	359	0.035	0.015	-0.020	0.000	0.364	0.568
budget 032 full	401	0.030	0.018	-0.012	0.000	0.374	0.573
budget 032 full	463	0.033	0.016	-0.016	0.000	0.407	0.557
budget 032 full	509	0.030	0.019	-0.012	0.000	0.381	0.604
budget 032 full	571	0.031	0.018	-0.013	0.000	0.397	0.599
budget 032 full	631	0.031	0.017	-0.014	0.000	0.388	0.562
budget 032 full	701	0.032	0.018	-0.014	0.000	0.406	0.594
budget 032 full	761	0.026	0.015	-0.011	0.000	0.413	0.651
budget 032 full	823	0.027	0.011	-0.015	0.000	0.392	0.615
budget 032 full	887	0.031	0.015	-0.017	0.000	0.360	0.609
budget 032 full	947	0.033	0.017	-0.016	0.000	0.372	0.599
budget 128 full	11	0.029	0.006	-0.023	0.000	0.392	0.573
budget 128 full	29	0.028	0.007	-0.021	0.000	0.400	0.620
budget 128 full	47	0.029	0.006	-0.023	0.000	0.385	0.609

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
budget 128 full	83	0.027	0.006	-0.021	0.000	0.409	0.630
budget 128 full	131	0.028	0.006	-0.022	0.000	0.381	0.646
budget 128 full	173	0.036	0.005	-0.032	0.000	0.381	0.562
budget 128 full	211	0.028	0.008	-0.020	0.000	0.398	0.630
budget 128 full	257	0.034	0.006	-0.027	0.000	0.375	0.516
budget 128 full	307	0.029	0.008	-0.022	0.000	0.380	0.594
budget 128 full	359	0.035	0.005	-0.030	0.000	0.364	0.568
budget 128 full	401	0.030	0.006	-0.024	0.000	0.374	0.573
budget 128 full	463	0.033	0.006	-0.027	0.000	0.407	0.557
budget 128 full	509	0.030	0.005	-0.025	0.000	0.381	0.604
budget 128 full	571	0.031	0.008	-0.023	0.000	0.397	0.599
budget 128 full	631	0.031	0.008	-0.023	0.000	0.388	0.562
budget 128 full	701	0.032	0.006	-0.026	0.000	0.406	0.594
budget 128 full	761	0.026	0.006	-0.020	0.000	0.413	0.651
budget 128 full	823	0.027	0.004	-0.023	0.000	0.392	0.615
budget 128 full	887	0.031	0.006	-0.025	0.000	0.360	0.609
budget 128 full	947	0.033	0.006	-0.027	0.000	0.372	0.599
obs full	11	0.029	0.010	-0.019	0.000	0.392	0.573
obs full	29	0.028	0.009	-0.019	0.000	0.400	0.620
obs full	47	0.029	0.008	-0.021	0.000	0.385	0.609
obs full	83	0.027	0.007	-0.020	0.000	0.409	0.630
obs full	131	0.028	0.009	-0.019	0.000	0.381	0.646
obs full	173	0.036	0.005	-0.031	0.000	0.381	0.562
obs full	211	0.028	0.010	-0.018	0.000	0.398	0.630
obs full	257	0.034	0.009	-0.024	0.000	0.375	0.516
obs full	307	0.029	0.008	-0.021	0.000	0.380	0.594
obs full	359	0.035	0.008	-0.027	0.000	0.364	0.568
obs full	401	0.030	0.012	-0.018	0.000	0.374	0.573
obs full	463	0.033	0.009	-0.024	0.000	0.407	0.557
obs full	509	0.030	0.007	-0.024	0.000	0.381	0.604
obs full	571	0.031	0.011	-0.020	0.000	0.397	0.599
obs full	631	0.031	0.010	-0.021	0.000	0.388	0.562
obs full	701	0.032	0.011	-0.021	0.000	0.406	0.594
obs full	761	0.026	0.008	-0.018	0.000	0.413	0.651
obs full	823	0.027	0.007	-0.020	0.000	0.392	0.615
obs full	887	0.031	0.007	-0.024	0.000	0.360	0.609
obs full	947	0.033	0.008	-0.024	0.000	0.372	0.599
obs coarse	11	0.029	0.017	-0.012	0.000	0.440	0.568
obs coarse	29	0.029	0.017	-0.012	0.000	0.453	0.630
obs coarse	47	0.025	0.014	-0.010	0.000	0.453	0.646
obs coarse	83	0.031	0.014	-0.017	0.000	0.456	0.578
obs coarse	131	0.031	0.017	-0.014	0.000	0.434	0.615
obs coarse	173	0.031	0.016	-0.014	0.000	0.446	0.609
obs coarse	211	0.028	0.015	-0.013	0.000	0.453	0.630
obs coarse	257	0.033	0.016	-0.017	0.000	0.441	0.547
obs coarse	307	0.028	0.016	-0.012	0.000	0.452	0.620
obs coarse	359	0.031	0.015	-0.016	0.000	0.461	0.599
obs coarse	401	0.029	0.014	-0.015	0.000	0.443	0.625
obs coarse	463	0.029	0.017	-0.012	0.000	0.457	0.604
obs coarse	509	0.030	0.015	-0.016	0.000	0.441	0.635
obs coarse	571	0.029	0.014	-0.015	0.000	0.464	0.625
obs coarse	631	0.033	0.016	-0.017	0.000	0.433	0.557
obs coarse	701	0.026	0.015	-0.012	0.000	0.477	0.656
obs coarse	761	0.028	0.016	-0.012	0.000	0.464	0.620
obs coarse	823	0.032	0.016	-0.016	0.000	0.445	0.573
obs coarse	887	0.033	0.017	-0.017	0.000	0.425	0.594
obs coarse	947	0.032	0.014	-0.018	0.000	0.454	0.609
obs blind	11	0.029	0.028	-0.001	0.000	0.442	0.568
obs blind	29	0.029	0.025	-0.004	0.000	0.444	0.630
obs blind	47	0.025	0.023	-0.002	0.000	0.454	0.651
obs blind	83	0.030	0.026	-0.004	0.000	0.457	0.589
obs blind	131	0.031	0.027	-0.003	0.000	0.426	0.615

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
obs blind	173	0.032	0.028	-0.004	0.000	0.442	0.599
obs blind	211	0.029	0.025	-0.005	0.000	0.450	0.615
obs blind	257	0.033	0.029	-0.003	0.000	0.435	0.542
obs blind	307	0.029	0.027	-0.002	0.000	0.441	0.615
obs blind	359	0.031	0.025	-0.006	0.000	0.444	0.604
obs blind	401	0.030	0.027	-0.003	0.000	0.437	0.620
obs blind	463	0.029	0.026	-0.003	0.000	0.451	0.594
obs blind	509	0.030	0.025	-0.005	0.000	0.436	0.641
obs blind	571	0.030	0.025	-0.005	0.000	0.454	0.620
obs blind	631	0.033	0.029	-0.005	0.000	0.426	0.562
obs blind	701	0.027	0.024	-0.004	0.000	0.470	0.641
obs blind	761	0.028	0.024	-0.004	0.000	0.468	0.630
obs blind	823	0.032	0.026	-0.006	0.000	0.435	0.562
obs blind	887	0.032	0.028	-0.004	0.000	0.423	0.604
obs blind	947	0.031	0.027	-0.004	0.000	0.449	0.604
blind undercov.	11	0.051	0.055	0.004	0.000	0.313	0.398
blind undercov.	29	0.053	0.052	-0.002	0.000	0.324	0.414
blind undercov.	47	0.048	0.052	0.004	0.000	0.330	0.434
blind undercov.	83	0.047	0.053	0.006	0.000	0.357	0.430
blind undercov.	131	0.049	0.054	0.004	0.000	0.315	0.434
blind undercov.	173	0.048	0.049	0.001	0.000	0.316	0.469
blind undercov.	211	0.049	0.050	0.001	0.000	0.327	0.461
blind undercov.	257	0.051	0.055	0.004	0.000	0.309	0.406
blind undercov.	307	0.043	0.046	0.003	0.000	0.339	0.477
blind undercov.	359	0.050	0.057	0.007	0.000	0.315	0.426
blind undercov.	401	0.047	0.053	0.005	0.000	0.334	0.398
blind undercov.	463	0.049	0.051	0.002	0.000	0.352	0.391
blind undercov.	509	0.048	0.050	0.002	0.000	0.348	0.434
blind undercov.	571	0.051	0.052	0.001	0.000	0.362	0.418
blind undercov.	631	0.053	0.055	0.002	0.000	0.297	0.418
blind undercov.	701	0.051	0.054	0.003	0.000	0.323	0.430
blind undercov.	761	0.051	0.057	0.006	0.000	0.316	0.438
blind undercov.	823	0.045	0.049	0.004	0.000	0.332	0.449
blind undercov.	887	0.048	0.051	0.003	0.000	0.343	0.426
blind undercov.	947	0.048	0.054	0.006	0.000	0.322	0.449
sparse group04	11	0.021	0.007	-0.014	0.000	0.407	0.486
sparse group04	29	0.023	0.007	-0.016	0.000	0.442	0.528
sparse group04	47	0.020	0.007	-0.013	0.000	0.413	0.472
sparse group04	83	0.024	0.007	-0.017	0.000	0.432	0.458
sparse group04	131	0.019	0.006	-0.013	0.000	0.433	0.479
sparse group04	173	0.024	0.006	-0.018	0.000	0.399	0.472
sparse group04	211	0.026	0.008	-0.018	0.000	0.404	0.493
sparse group04	257	0.025	0.007	-0.018	0.000	0.419	0.431
sparse group04	307	0.025	0.007	-0.018	0.000	0.412	0.403
sparse group04	359	0.022	0.008	-0.014	0.000	0.412	0.493
sparse group04	401	0.023	0.008	-0.014	0.000	0.417	0.479
sparse group04	463	0.020	0.005	-0.014	0.000	0.435	0.507
sparse group04	509	0.024	0.007	-0.017	0.000	0.439	0.424
sparse group04	571	0.027	0.009	-0.018	0.000	0.408	0.472
sparse group04	631	0.022	0.007	-0.015	0.000	0.399	0.451
sparse group04	701	0.023	0.007	-0.015	0.000	0.435	0.472
sparse group04	761	0.022	0.009	-0.013	0.000	0.429	0.493
sparse group04	823	0.026	0.005	-0.021	0.000	0.410	0.500
sparse group04	887	0.025	0.007	-0.018	0.000	0.412	0.479
sparse group04	947	0.022	0.008	-0.015	0.000	0.385	0.500
sparse group08	11	0.023	0.004	-0.019	0.000	0.409	0.497
sparse group08	29	0.024	0.005	-0.019	0.000	0.434	0.455
sparse group08	47	0.021	0.005	-0.016	0.000	0.425	0.462
sparse group08	83	0.022	0.004	-0.018	0.000	0.417	0.455
sparse group08	131	0.022	0.003	-0.019	0.000	0.430	0.458
sparse group08	173	0.022	0.004	-0.018	0.000	0.415	0.469
sparse group08	211	0.024	0.004	-0.020	0.000	0.405	0.455

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Table 16: Raw case-by-seed calibrated-error rows, continued.

Case	Seed	Group MSE	Critic MSE	Δ MSE	Zero-var	Wait	Success
sparse group08	257	0.023	0.005	-0.018	0.000	0.414	0.462
sparse group08	307	0.023	0.004	-0.019	0.000	0.419	0.417
sparse group08	359	0.022	0.005	-0.017	0.000	0.419	0.472
sparse group08	401	0.022	0.005	-0.017	0.000	0.417	0.479
sparse group08	463	0.022	0.003	-0.019	0.000	0.429	0.507
sparse group08	509	0.023	0.004	-0.019	0.000	0.428	0.476
sparse group08	571	0.024	0.004	-0.020	0.000	0.415	0.427
sparse group08	631	0.021	0.004	-0.017	0.000	0.417	0.465
sparse group08	701	0.021	0.004	-0.017	0.000	0.434	0.490
sparse group08	761	0.023	0.004	-0.019	0.000	0.434	0.479
sparse group08	823	0.024	0.005	-0.019	0.000	0.402	0.444
sparse group08	887	0.025	0.004	-0.021	0.000	0.419	0.476
sparse group08	947	0.020	0.004	-0.017	0.000	0.416	0.465

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G Reproducibility Checklist

1. Run the unit tests with `python3 -m unittest discover -s tests`.
2. Run the smoke gate with `./scripts/run_smoke.sh`.
3. Regenerate the deep matrix with `python3 -m experiments.deep_matrix`.
4. Regenerate LaTeX inputs with `python3 scripts/build_latex_inputs.py`.
5. Build the full paper with `./scripts/build_latex_paper.sh`.
6. Verify that the LaTeX artifact manifest records the PDF hash and page count.
7. Treat `public/trajectory_rewards_are_not_token_credit.pdf` as the single canonical paper PDF.

H Next Experiments

Full clipped PPO/GRPO training. The current closed-loop audit is tabular and unclipped. The next controlled RL experiment should implement full clipped PPO, response-level GRPO, and corrected GRPO variants such as length-normalized, leave-one-out, DAPO-style, and Dr. GRPO-style updates under equal generated-token and optimizer budgets. This will test whether the estimator-fidelity and tabular-policy findings survive the optimizer details used in modern post-training stacks.

Tiny sequence-policy benchmark. The strongest next upgrade is a small transformer or nanoGPT-style benchmark where the environment remains controlled but the policy, critic, and reward model consume token histories rather than compact toy features. The v0.9 neural audit checks value interpolation on a held-out toy threshold; it does not yet test sequence modeling, KL control, or clipped policy optimization.

Process rewards. Adding process labels or subgoal rewards should narrow the gap between critic-based and group-relative estimators. This prediction is important because many practical systems do not rely on terminal reward alone.

Compaction. A compaction experiment should split long traces into variable numbers of sub-traces. The hypothesis is that critic-based individual-rollout objectives handle this naturally, while group-wise objectives need additional rules for matching sub-traces across completions.

Distillation hybrids. A dense-distillation follow-up should use successful RL traces as teacher or privileged self-teacher material and measure whether dense distillation preserves the credit assignment gains at lower inference cost.

I Replication Protocol for New Benchmarks

Trace schema. A real long-horizon benchmark should record more than a final reward. Each trajectory should include prompt identity, sampled response identity, token or step indices, tool calls, tool observations, compaction boundaries, verifier calls, process labels when available, terminal reward, and any safety or refusal events. This schema is the minimum needed to compare a group-relative estimator with a value-model estimator after the fact. Without it, the experiment can still report final success, but it cannot diagnose where credit was assigned or lost.

Budget matching. The clean comparison is not “PPO run” versus “GRPO run” in isolation. The comparison should state generated-token budget, number of prompts, completions per prompt, verifier calls, critic update steps, policy update steps, and peak activation or optimizer memory. A critic-based method should be charged for critic fitting and storage. A group-relative method should be charged for the extra completions needed to obtain reward contrast. A distillation method should be charged for teacher inference or self-teacher selection.

Primary measurements. For any benchmark with exact or process annotations, report estimator correlation with behavior-policy credit, calibrated MSE, sign accuracy, zero-variance group fraction, and return improvement under equal rollout budgets. For benchmarks without exact credit, report held-out process-label agreement, per-step value calibration where labels exist, and sensitivity to trajectory length. Final score alone is not enough because a method can improve a short task while failing the mechanism required for long traces.

Decision rule. A public claim that one family is better should require at least three checks: the mean effect favors the method, the confidence interval excludes practical near-ties, and an adversarial or counterexample regime is reported. This paper’s toy result follows that rule: it highlights the critic-favorable mean pattern, the near tie under tiny critic budget, and the group-favorable blind undercovered case. A larger benchmark should preserve the same discipline.

Audit package. The public release should include raw per-seed rows, aggregate tables, plotting scripts, exact command lines, environment notes, and a short statement of non-claims. The non-claims matter. For this workstream, the honest claim is that critic-style temporal information can dominate in observable long traces, while group-relative normalization remains strong when reward contrast is reliable and value modeling is weak. That is a research program, not a slogan.

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